

MY MONEY: Transaction Management Application

A PROJECT REPORT

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in partial fulfillment for the award of the degree

of

BACHELORS OF COMPUTER APPLICATIONS

IN

DEPARTMENT OF COMPUTATIONAL SCIENCES

BRAINWARE UNIVERSITY

398, Ramkrishnapur Road, Barasat, North 24 Parganas, Kolkata - 700 125



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ABSTRACT

The rapid growth of digital technologies and the widespread use of smartphones have significantly transformed how individuals manage their personal finances. Traditional methods of financial tracking are increasingly being replaced by mobile applications that offer convenience, accessibility, and real-time insights. This project, titled “MY MONEY: Transaction Management System,” aims to develop a comprehensive offline-first mobile application that enables users to manage their financial activities efficiently and securely. The system integrates essential features such as transaction recording, budget categorization, relational expense tracking, analytical dashboards, and automated financial operations. By consolidating these functionalities into a single platform, the application provides a structured and user-friendly financial management experience.

The project follows a modern mobile application development approach using React Native with the Expo framework, SQLite for local data persistence, and state management techniques for efficient data handling. The application is designed to operate entirely offline, ensuring user privacy and eliminating dependency on external servers or cloud-based systems. The backend logic is managed locally through structured database interactions and optimized queries, while the frontend delivers a responsive and intuitive interface for seamless user interaction. The system also incorporates advanced features such as relational ledger management, dynamic data visualization using SVG-based graphs, and background automation processes for scheduled transactions.

Through detailed requirement analysis, modular system design, implementation, and testing, the MY MONEY application demonstrates the effective use of contemporary mobile technologies in the domain of personal finance management. Experimental results confirm the reliability, performance, and usability of the system under various scenarios. The application establishes a strong foundation for future enhancements such as secure data synchronization, predictive financial analytics, and advanced reporting tools. Overall, the project presents a reliable, privacy-focused, and extensible financial management solution suitable for individual users seeking efficient control over their financial activities.

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LIST OF SYMBOLS, ABBREVIATIONS, AND NOMENCLATURE

Symbol / Abbreviation	Meaning
UI	User Interface
UX	User Experience
API	Application Programming Interface
DB	Database
JSON	JavaScript Object Notation
UUID	Universally Unique Identifier
PK	Primary Key
FK	Foreign Key
CLD	Context Level Diagram
DFD	Data Flow Diagram
ERD	Entity–Relationship Diagram
AutoPay	Automatic recurring transactions
Ledger	Record of all financial transactions
Offline-first	System works without internet dependency
Cloud Sync	Data synchronization across devices
AI	Artificial Intelligence
OCR	Optical Character Recognition
P2P	Peer-to-Peer transactions

Symbol / Abbreviation**Meaning**

NPCI

National Payments Corporation of India

RBI

Reserve Bank of India

UPI

Unified Payments Interface

UCD

Use Case Diagram

SQLite

Lightweight relational database system

React Native

Framework for mobile app development

Zustand

Lightweight relational database system



CLD External Entities



CLD System



CLD Unidirectional Flow



DFD External Entity



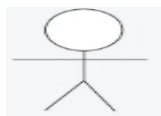
DFD Process



DFD Data Flow



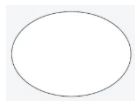
DFD Data Store



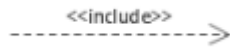
Use Case Diagram Actor

Symbol / Abbreviation

Meaning



Use Case Diagram Usecase



Use Case Diagram Include Relation



Use Case Diagram Communication Line



ERD Entity



ERD One and only one Cardinality



ERD Zero or Many Cardinality



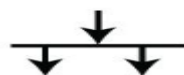
Activity Diagram Initial State



Activity Diagram Final State



Activity Diagram Decision Node



Activity Diagram Fork



Activity Diagram Activity State



Flowchart Process



Flowchart Decision



Flowchart Connector

CHAPTER 1

INTRODUCTION

The rapid advancement of digital technologies, coupled with the widespread adoption of smartphones, has significantly transformed how individuals manage their daily activities, including personal finance. In recent years, mobile applications have emerged as powerful tools for simplifying complex tasks such as budgeting, expense tracking, and financial planning. Traditional methods of managing finances, such as manual record-keeping or spreadsheet-based tracking, are gradually being replaced by intelligent digital solutions that offer convenience, accuracy, and real-time insights. As users increasingly rely on mobile platforms for managing financial data, there is a growing demand for secure, efficient, and user-friendly applications that can operate seamlessly across different environments.

The project titled “MY MONEY: Transaction Management System” is designed to address this growing demand by providing a comprehensive mobile-based financial management solution. Unlike many existing applications that rely heavily on cloud-based services and external APIs, this system adopts an offline-first approach, ensuring that all user data is stored and processed locally on the device. This approach enhances user privacy, eliminates dependency on internet connectivity, and ensures consistent performance even in low-network or offline conditions. The application is developed using modern mobile technologies, including React Native with the Expo framework, SQLite for local database management, and advanced state management techniques to maintain efficient data flow within the system.

In the modern financial ecosystem, users often face challenges related to data privacy, performance limitations, and fragmented financial tools. Many popular financial applications require users to connect their bank accounts through third-party services, which can lead to potential privacy risks and dependency on external systems. Additionally, several applications separate core financial tracking from expense-sharing functionalities, forcing users to rely on multiple platforms to manage their finances effectively. These limitations create a need for a unified system that not only tracks financial transactions but also integrates relational aspects such as shared expenses, group transactions, and individual contributions within a single platform.

The MY MONEY application is designed to overcome these challenges by introducing a relational transaction management system that connects financial data with associated entities such as individuals and groups. This allows users to track not only how much they spend, but also with whom and where those financial interactions occur. The system supports multiple types of transactions, including income, expenses, and transfers, while maintaining a structured and organized database that ensures data consistency and accuracy. Through the

use of relational data models, the application provides a more comprehensive understanding of financial activities compared to traditional linear tracking systems.

Another key motivation behind this project is the need for enhanced performance and responsiveness in financial applications. Many existing systems rely on remote servers for data processing, which can lead to delays, synchronization issues, and inconsistent user experiences. In contrast, the MY MONEY application leverages a local SQLite database to perform all data operations directly on the device. This enables faster query execution, efficient handling of large datasets, and smooth interaction with the user interface. The system is capable of processing thousands of transaction records with minimal latency, ensuring a responsive and reliable user experience.

The application also incorporates advanced visualization techniques to help users better understand their financial patterns. By utilizing vector-based graphical representations, the system presents analytical data in the form of charts and graphs that dynamically update based on user interactions. These visualizations provide insights into spending habits, budget distribution, and financial trends over time. The integration of such features enhances the usability of the application and allows users to make informed financial decisions based on accurate and easily interpretable data.

The scope of the MY MONEY application extends across multiple functional areas, including transaction management, budget categorization, relational expense tracking, automated financial operations, and analytical reporting. The system is designed to support individual users who wish to maintain detailed records of their financial activities, as well as users who need to manage shared expenses within groups or among peers. Additionally, the application includes features such as background automation for scheduled transactions, ensuring that recurring financial activities are handled efficiently without requiring constant user intervention.

The conceptual scope of the system is summarized in Table 1.1, which outlines the different user types and their corresponding functionalities within the system.

Table 1.1 Scope of MY MONEY Modules

User Type	Functionalities
User	Record transactions, manage budgets, track expenses, view analytics
System	Local database management, data processing, visualization, automation

The MY MONEY application is built upon a structured database design that includes multiple interconnected entities such as accounts, transactions, categories, budgets, people, and groups. These entities are linked through relational mappings that ensure data integrity and enable complex financial operations to be executed efficiently. The use of SQLite as the underlying database system provides a lightweight yet powerful solution for managing structured data within a mobile environment.

From an academic perspective, this project demonstrates the practical application of mobile development concepts, database management techniques, and system design principles. It

reflects an understanding of how different components of a software system interact with each other to deliver a cohesive and functional application. The project also highlights the importance of selecting appropriate technologies based on system requirements, such as choosing local storage for enhanced privacy and performance. From a technical standpoint, the system emphasizes key aspects such as security, performance optimization, and modular architecture. By storing all data locally, the application ensures that sensitive financial information remains within the user's device, reducing the risk of unauthorized access. Efficient database queries and optimized state management contribute to improved performance, while the modular design allows for easy maintenance and future enhancements.

Furthermore, the application focuses on delivering a seamless user experience by maintaining consistency in design, navigation, and interaction patterns. The user interface is designed to be intuitive and responsive, enabling users to perform tasks such as adding transactions, viewing reports, and managing budgets with minimal effort. Attention to detail in layout, spacing, and visual hierarchy ensures that users can interact with the application comfortably and efficiently.

In the broader context, personal finance management has become increasingly important as individuals seek to gain better control over their financial resources. A well-designed financial application can help users develop disciplined spending habits, plan budgets effectively, and achieve long-term financial goals. The MY MONEY application contributes to this objective by providing a reliable and accessible platform that simplifies financial management while maintaining high standards of privacy and performance.

The motivation for developing this system also stems from the need to create a self-contained financial solution that does not rely on external dependencies. By eliminating the need for cloud synchronization and third-party integrations, the application ensures greater control over data and reduces potential vulnerabilities associated with external services. This approach aligns with the growing demand for privacy-focused applications in the modern digital landscape.

In conclusion, this chapter has provided an overview of the MY MONEY: Transaction Management System, highlighting its purpose, features, scope, and significance. It has discussed the challenges associated with existing financial applications and explained how the proposed system addresses these issues through an offline-first, relational, and performance-oriented approach. The chapter establishes the foundation for understanding the subsequent sections of this report, which will further explore the objectives, design, implementation, and evaluation of the system.

The next chapter presents the detailed objectives that guided the development of the MY MONEY application.

CHAPTER 2

OBJECTIVE

The MY MONEY: Transaction Management System has been developed with a clearly defined set of objectives that guide the overall design, implementation, and functionality of the application. These objectives are framed to address the growing need for efficient, secure, and user-friendly financial management solutions in a mobile environment. In today's digital world, individuals rely heavily on mobile applications to manage their financial activities, making it essential to design systems that are reliable, responsive, and capable of operating independently without external dependencies. The objectives of this project ensure that the application not only performs essential financial operations but also provides enhanced privacy, performance, and usability. Unlike many existing financial applications that depend on cloud-based services, this system adopts an offline-first approach, where all financial data is stored and processed locally on the device. This approach ensures data security, eliminates dependency on internet connectivity, and provides consistent performance under all conditions. The objectives of the system are divided into different categories, including primary, secondary, functional, technical, user-centric, administrative, and long-term objectives. Each category plays a crucial role in shaping the system and ensuring that it meets both user expectations and technical standards.

2.1 Primary Objective

The primary objective of the MY MONEY application is to develop a comprehensive mobile-based transaction management system that enables users to efficiently record, organize, and analyze their financial activities. The system aims to provide a structured platform where users can manage income, expenses, and transfers while maintaining complete control over their data. This objective is achieved through the use of modern mobile development technologies such as React Native and Expo, along with SQLite for local database management. The system ensures fast performance, accurate data handling, and smooth user interaction. By providing a unified platform, the application eliminates the need for multiple tools and simplifies financial management for users.

2.2 Secondary Objectives

In addition to the primary objective, several secondary objectives guide the development of the system. These objectives focus on enhancing functionality, improving usability, and extending the capabilities of the application beyond basic transaction tracking. The system

aims to provide a complete offline solution that allows users to access all features without internet connectivity. It ensures data privacy by storing all financial information locally on the device, reducing the risk of unauthorized access. Another important objective is to introduce relational financial tracking, allowing users to associate transactions with individuals and groups. This helps in managing shared expenses and provides better insight into financial interactions. The application also focuses on presenting financial data in a clear and meaningful way. By incorporating visual analytics such as charts and summaries, users can easily understand their financial patterns. Additionally, automation features such as scheduled transactions help reduce manual effort and improve efficiency in managing recurring financial activities.

2.3 Functional Objectives

The functional objectives describe the key operations that the system must perform. These objectives ensure that the application provides all essential features required for effective financial management. The detailed functional objectives of the MY MONEY application are presented in Table 2.1.

Table 2.1 Functional Objectives of MY MONEY Application

Area	Functional Objective
Transaction Management	Record income, expenses, and transfer transactions
Budget Management	Create and manage budget categories
Relational Tracking	Track shared expenses with people and groups
Analytics Dashboard	Display financial summaries and insights
Automation	Handle recurring and scheduled transactions
Data Storage	Store all data securely in a local database

2.4 Technical Objectives

The technical objectives focus on the technologies and architectural design that ensure system performance, reliability, and scalability. These objectives highlight the importance of using modern tools and efficient system design practices. The system is developed using React Native with Expo to ensure cross-platform mobile compatibility. SQLite is used as the primary database to provide efficient local data storage and fast query execution. The application follows a modular architecture that separates the user interface, business logic, and database layers, making the system easier to maintain and scale. Another important objective is to ensure high performance by optimizing database queries and minimizing response time. The system is designed to handle large volumes of transaction data efficiently without affecting the user experience.

2.5 User-centric Objectives

The user-centric objectives focus on improving the overall user experience and ensuring that the application remains simple and easy to use. The system is designed to provide a clean and intuitive interface that allows users to perform financial operations without confusion. Users should be able to add, edit, and manage transactions quickly and efficiently. Financial data should be presented clearly so that users can easily understand their financial status. The application also ensures smooth navigation across different sections, reducing complexity and improving usability.

2.6 Administrative Objectives

The administrative objectives ensure proper management and organization of financial data within the system. These objectives help maintain system reliability and consistency. The system ensures that all financial records are stored accurately and consistently. It organizes data into structured formats, making it easier to retrieve and analyze information. Automated processes such as recurring transactions are handled efficiently to reduce manual intervention. Overall system stability is maintained to ensure uninterrupted operation.

2.7 Long-term Objectives

The long-term objectives focus on future improvements and scalability of the system. These objectives ensure that the application can adapt to evolving technologies and user needs. Future enhancements may include secure data synchronization across multiple devices, allowing users to access their financial data from different platforms. The system can also be expanded to include AI-based financial analysis, providing predictive insights and recommendations. Additional features such as report generation and advanced visualization tools can further improve the functionality of the application. The modular design of the system ensures that new features can be added without major changes to the existing architecture.

This chapter presented the objectives of the MY MONEY: Transaction Management System. The primary and secondary objectives defined the overall purpose of the system, while the functional, technical, user-centric, administrative, and long-term objectives explained how the application meets user requirements and technical standards. These objectives provide a clear direction for the development of the system and ensure that it remains efficient, secure, and scalable. The next chapter will focus on Planning, which includes the development methodology, project scheduling, module identification, resource allocation, and risk analysis.

CHAPTER 3

PLANNING

Effective planning is a crucial phase in the development of any software system, especially for a mobile-based application like the MY MONEY: Transaction Management System. Proper planning ensures that the development process follows a structured approach, reduces complexity, and enables efficient utilization of resources. It also helps in identifying potential challenges at an early stage and provides strategies to overcome them. For this project, planning played a significant role in organizing tasks, selecting appropriate technologies, and ensuring timely completion of all modules. The planning phase involved requirement understanding, selection of development methodology, identification of system modules, scheduling of tasks, resource allocation, and risk management. By following a systematic planning strategy, the project ensures consistency, reliability, and smooth execution of development activities.

3.1 Need for Planning

Planning is essential in the development of the MY MONEY: Transaction Management System as it helps manage the complexity of building a mobile application with multiple features such as transaction tracking, analytics, and automation. Proper planning ensures that all components of the system, including the user interface, application logic, and database, are well-organized and work together efficiently. It also helps in selecting appropriate technologies, managing time effectively, and dividing the project into manageable phases. Additionally, planning allows early identification of potential risks and challenges, ensuring smooth development and successful completion of the project.

3.2 Project Development Methodology

The MY MONEY application follows a modified Agile development methodology. Agile is suitable for this project because it supports iterative and incremental development, allowing continuous improvement throughout the development lifecycle. The development process is divided into multiple phases, where each phase focuses on a specific module. After each phase, testing and refinement are performed to ensure correctness and performance. This approach provides flexibility and allows changes to be incorporated easily during development. The overall development methodology is summarized in Table 3.1.

Table 3.1 Development Methodology Overview

Phase	Description
Requirement Analysis	Understanding user needs and system requirements
Planning	Defining modules, tools, and workflow
Design	Creating system architecture and UI layout
Implementation	Developing application features
Testing	Verifying functionality and performance
Deployment	Preparing the final application
Review	Improving system based on feedback

3.3 Work Breakdown Structure (WBS)

The Work Breakdown Structure (WBS) for the MY MONEY application serves as a foundational framework that divides the project into smaller, manageable tasks to ensure systematic development and execution. The application is organized into several key components, starting with Frontend Development, which centers on designing intuitive UI screens, implementing seamless navigation, and handling user interactions. This interface is powered by the Application Logic, which manages core functionalities such as transaction processing, rigorous data validation, and automation handling. Complementing this is the Database Design, which involves creating structured tables using SQLite, defining complex relationships between data entities, and managing secure, reliable data storage.

To derive value from this data, the Analytics & Visualization component is responsible for processing financial information to generate actionable insights, charts, and reports. Efficiency is further enhanced by the Automation Module, which handles essential background processing tasks and the scheduling of recurring financial transactions. Finally, the Testing & Debugging component ensures overall system reliability by rigorously validating expected behaviors and systematically addressing bugs and errors. By breaking the project into these distinct modules, the WBS facilitates improved task management and enables the smooth, coordinated integration of all system features.

3.4 Project Scheduling

Project scheduling ensures that all development activities are completed within a defined timeframe. It helps in tracking progress and maintaining consistency throughout the development process. The detailed project schedule is presented in Table 3.2.

Table 3.2 Project Schedule (Gantt Chart)

Activity	1	2	3	4	5	6	7	8	9	10	11	12
Requirement analysis												
UI design & setup												

Activity	1	2	3	4	5	6	7	8	9	10	11	12
Transaction management												
Database design												
Analytics & dashboard												
Automation features												
Testing & debugging												
Optimization												
Final review												

3.5 Module Identification

Module identification is an important step in planning as it helps divide the system into functional units. Each module is responsible for a specific set of operations and interacts with other modules.

The main modules of the MY MONEY application are:

- Transaction Module (handles income, expenses, transfers)
- Budget Module (manages categories and limits)
- Relational Module (handles shared expenses)
- Analytics Module (provides financial insights)
- Automation Module (handles recurring transactions)

This modular approach reduces complexity and improves system maintainability.

3.6 Resource Planning

Resource planning is a critical phase of the MY MONEY project, involving the systematic identification of all technical and human resources necessary to ensure efficient utilization and project success. On the technical side, the project relies on Visual Studio Code as the primary development environment, alongside a robust technology stack consisting of React Native, Expo, and SQLite. These are further supplemented by specialized libraries dedicated to state management and UI component rendering. Complementing these technical assets are the human resources, which include the developers responsible for building the system, a UI/UX designer to craft the interface, a tester to ensure quality assurance, and a documentation writer to maintain project records. By strategically aligning these tools and

personnel, the project achieves an optimized workflow, ensuring that both manpower and technological capabilities are used effectively throughout the development lifecycle.

3.7 Risk Analysis and Mitigation

Risk analysis is a vital component of the MY MONEY development lifecycle, as it proactively identifies potential challenges and establishes targeted strategies to minimize their impact on the final system. To address performance issues related to handling large volumes of transaction data, the project team will focus on optimizing database queries to ensure rapid retrieval and processing. Navigating the complexities of relational tracking is managed through the implementation of a strictly structured database design, which simplifies data management. Furthermore, to mitigate the risks associated with UI complexity, the project prioritizes a simple, intuitive, and consistent interface to maintain a positive user experience. Finally, to handle potential system errors, the development process includes a rigorous, ongoing cycle of regular testing and debugging to catch and resolve bugs early. By identifying and addressing these risks at an early stage, the project ensures that the application maintains stable, efficient, and reliable performance throughout its operation.

This chapter presented the planning approach followed for the development of the MY MONEY: Transaction Management System. It discussed the importance of planning, the development methodology used, work breakdown structure, project scheduling, module identification, resource planning, and risk analysis. Proper planning ensured that the system was developed in a structured and efficient manner, reducing complexity and improving overall system quality. The next chapter will focus on Requirement Analysis, which defines the system requirements, functionalities, and constraints in detail.

CHAPTER 4

REQUIREMENT ANALYSIS

Requirement analysis is one of the most important phases in the development of the MY MONEY: Transaction Management System. It defines what the system should do, how users will interact with it, and what technical conditions must be satisfied for proper functioning. This phase ensures that the application is developed according to user expectations and system requirements while maintaining performance, security, and usability.

The MY MONEY application is designed as a mobile-based, offline-first system that manages personal financial data efficiently. Therefore, requirement analysis plays a crucial role in identifying user needs, defining system functionalities, and ensuring that all components work together seamlessly. This chapter presents the user requirements, functional requirements, non-functional requirements, system requirements, data requirements, constraints, and assumptions of the system.

4.1 User Requirements

User requirements define the expectations and needs of the users interacting with the system. The MY MONEY application is primarily designed for individual users who want to manage their financial activities efficiently. Users expect a simple and intuitive interface that allows them to easily record transactions, manage budgets, and view financial summaries. They require a system that works without internet connectivity and ensures complete privacy of their financial data. The ability to track expenses, categorize transactions, and analyse spending patterns is essential for effective financial management. Additionally, users need features that allow them to manage shared expenses with friends or groups. They also expect automation features such as recurring transactions to reduce manual effort. Overall, the system must provide a smooth, fast, and secure user experience.

4.2 Functional Requirements

Functional requirements describe the operations that the system must perform. These requirements ensure that all essential features of the MY MONEY application are implemented correctly. The detailed functional requirements are presented in Table 4.1.

Table 4.1 Functional Requirements

Requirement No.	Description
FR-01	The system must allow users to record income, expenses, and transfer transactions
FR-02	The system must support creation and management of budget categories
FR-03	The system must allow users to track shared expenses with individuals and groups
FR-04	The system must provide graphical representation of financial data
FR-05	The system must support recurring and scheduled transactions
FR-06	The system must store all data securely in a local database
FR-07	The system must allow users to edit and delete transactions
FR-08	The system must provide real-time updates of financial data

4.3 Non-Functional Requirements

The non-functional requirements define the performance, security, usability, and reliability standards of the system, as illustrated in Table 4.2.

Table 4.2 Non-Functional Requirements

Requirement No.	Description
NFR-01	High performance and fast execution for all operations
NFR-02	Efficient data retrieval from large local transaction sets
NFR-03	Local data storage to ensure security and network isolation
NFR-04	Simple and intuitive user interface design
NFR-05	Reliable operation in offline environments
NFR-06	Scalable architecture for future feature expansion

4.4 System Requirements

The MY MONEY application requires a specific software environment to ensure seamless development and execution. The development process is supported by Visual Studio Code as the primary development tool, utilizing React Native with Expo as the core framework for building the application. To handle data storage, the system employs SQLite, while the codebase is developed using JavaScript or TypeScript. Once built, the application is designed to be fully compatible with both Android and iOS operating systems, ensuring broad accessibility and consistent performance across these platforms.

4.5 Data Requirements

The MY MONEY application uses structured data to store and manage financial information efficiently. The data includes user transactions, categories, budgets, accounts, and relational data involving people and groups. Each transaction record contains details such as type, amount, category, and date. The system also maintains relationships between transactions and other entities, enabling advanced features such as shared expense tracking. The database is designed using SQLite, which ensures efficient storage and fast retrieval of data. Proper data structuring ensures consistency, accuracy, and easy access to financial information within the application.

4.6 Existing Systems and Comparison

There are several financial management applications available in the market such as Walnut, Money View, and ET Money. These applications provide features like expense tracking, budgeting, and basic analytics. However, these systems have several limitations. Most of them rely on SMS-based detection or cloud synchronization, which can lead to incomplete data and privacy concerns. They also lack important features such as peer-to-peer debt tracking, group expense management, and accurate real-time balance computation. Additionally, many applications treat internal transfers incorrectly as expenses or do not support them at all.

The MY MONEY: Transaction Management System addresses these issues by providing a unified and offline-first solution. It offers complete manual account management with mathematically accurate balance calculation, ensuring data consistency. The system also includes advanced features such as peer-to-peer debt management, group expense clustering, real-time budget tracking, and autopay management. Furthermore, MY MONEY ensures full data privacy by storing all user data locally on the device without requiring server interaction. It also provides advanced multi-modal analytics and a one-time purchase model, making it more user-friendly and efficient compared to existing systems. A detailed comparison between existing systems and the MY MONEY application is presented in Table 4.3.

Table 4.3 Comparison of Existing Systems and MY MONEY

Capability	My Money	Walnut	Money View	ET Money
Manual Account Management	Full	SMS-inferred	Limited	Limited
Real-Time Account Balance	Mathematical	Approximate	Synced	None
Peer-to-Peer Debt Engine	Full	None	None	None
Group Expense Clustering	Full	None	None	None
Budget with Burn-Rate	Real-time	Basic	Basic	Basic
Monthly Budget Archival	Permanent	None	None	None

Autopay Management	Full	None	Reminder only	None
Historical Balance Reconstruction	Per-transaction	None	None	None
Internal Transfer Handling	Separate entity	Treated as expense	None	None
Offline-First Operation	Full	No	No	No
User Data Privacy	100% Local	SMS access	Server-stored	Server-stored
Advanced Multi-Modal Analytics	Full	Basic	Moderate	Investment-only
One-Time Purchase Model	Yes	Freemium/Ad	Subscription	Subscription
UPI Payment Integration (Roadmap)	Planned	None	None	None

4.7 Constraints

Constraints refer to limitations that affect the system during development and operation. One of the main constraints of the MY MONEY application is that it operates entirely offline, which means there is no real-time synchronization across multiple devices. While this improves privacy, it limits data sharing capabilities. Another constraint is related to mobile device performance, where the application must be optimized to run smoothly on devices with limited resources. Additionally, since the system uses local storage, users must manage backups manually if required. These constraints are considered during development to ensure the system remains efficient and reliable.

4.8 Assumptions

Assumptions define the conditions under which the system is expected to operate. It is assumed that users will use modern smartphones that support the application's features. Users are expected to enter accurate financial data to ensure correct analysis. It is also assumed that the device has sufficient storage to maintain transaction records. Another assumption is that users prefer privacy-focused applications and are comfortable managing their data locally without cloud-based synchronization.

This chapter discussed the requirement analysis of the MY MONEY: Transaction Management System, including functional and non-functional requirements, system specifications, and constraints. The next chapter will present the System Flow with various diagrams and flowcharts.

CHAPTER 5

SYSTEM FLOW

System flow describes how different components of the MY MONEY: Transaction Management System interact with each other to perform various operations. It provides a clear understanding of how data moves within the system, how users interact with different modules, and how the application processes and stores financial information. System flow diagrams help in visualizing the internal working of the system in a structured manner. The MY MONEY application is designed as an offline-first mobile system where all operations are performed locally on the device. The system accepts user inputs such as transactions, budgets, and relational data, processes them through application logic, and stores them in a structured SQLite database. The interaction between the user interface, system logic, and database is represented through different diagrams. This chapter includes the Context Level Diagram, Data Flow Diagrams, Use Case Diagram, Entity–Relationship Diagram, Activity Diagram, Sequence Diagram, and Flowcharts. Each diagram is explained briefly, and space is provided for inserting the figures.

5.1 Context Level Diagram (CLD)

The Context Level Diagram represents the MY MONEY system as a single entity interacting with external entities. It shows how users provide input to the system and receive output in the form of reports and analytics. The main external entity is the user, who interacts with the system by adding transactions, managing budgets, and viewing financial data. The system processes this information and provides outputs such as summaries and insights, as illustrated in Figure 5.1.

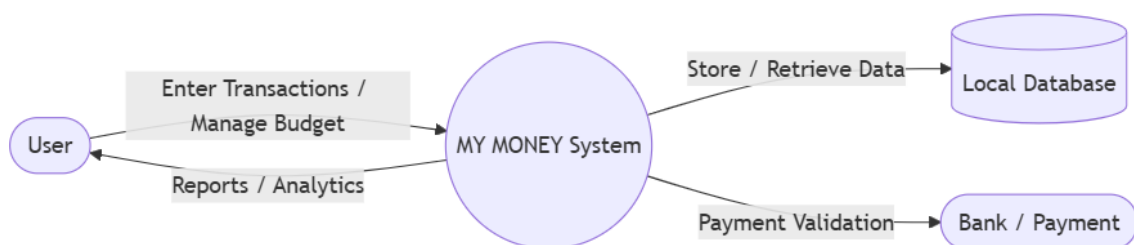


Figure 5.1: Context Level Diagram

5.2 Data Flow Diagram – Level 0 (DFD Level 0)

The Level 0 Data Flow Diagram provides an overview of the major processes within the system. It shows how data flows between the user, system processes, and the database. The main processes include transaction management, budget management, and analytics. The diagram illustrates how user inputs are processed and stored, and how results are generated, as illustrated in Figure 5.2.

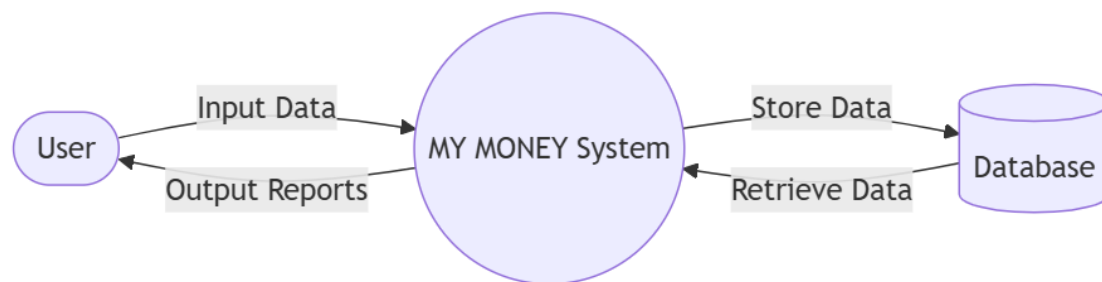


Figure 5.2: Data Flow Diagram Level 0

5.3 Data Flow Diagram – Level 1 (DFD Level 1)

The Level 1 Data Flow Diagram provides a more detailed representation of the system processes. It breaks down the main functions into smaller processes such as adding transactions, managing budgets, and viewing analytics. This diagram helps in understanding how individual components interact with the database and how data is processed step by step, as illustrated in Figure 5.3.

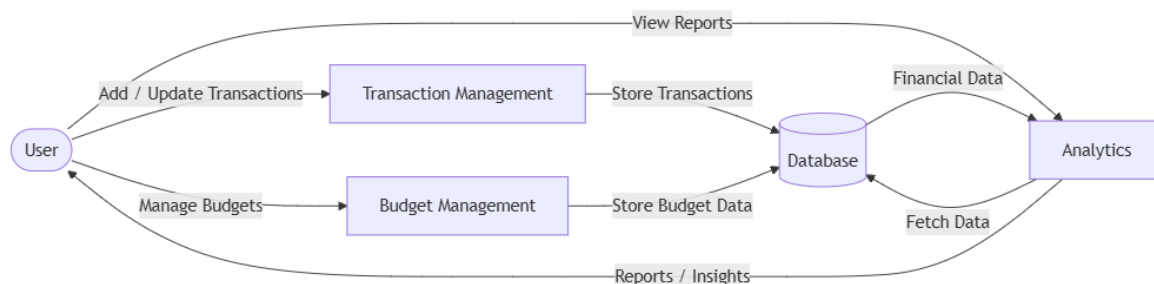


Figure 5.3: Data Flow Diagram Level 1

5.4 Data Flow Diagram – Level 2 (DFD Level 2)

The Level 2 Data Flow Diagram provides a detailed view of a specific process, such as transaction management. It shows how data is entered, validated, stored, and retrieved. This level of detail helps in understanding the internal working of the system and ensures accurate data processing, as illustrated in Figure 5.4.

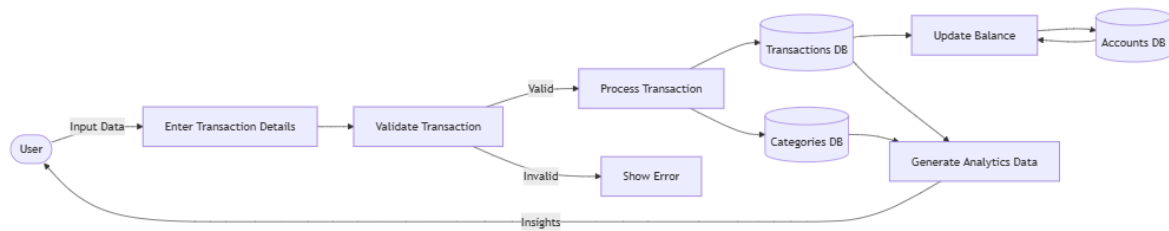


Figure 5.4: Data Flow Diagram Level 2

5.5 Use Case Diagram

The Use Case Diagram represents the interaction between the user and the system. It identifies the different operations that can be performed within the application. In the MY MONEY system, the user can perform actions such as adding transactions, managing budgets, viewing analytics, managing groups, and setting automation, as illustrated in Figure 5.5.

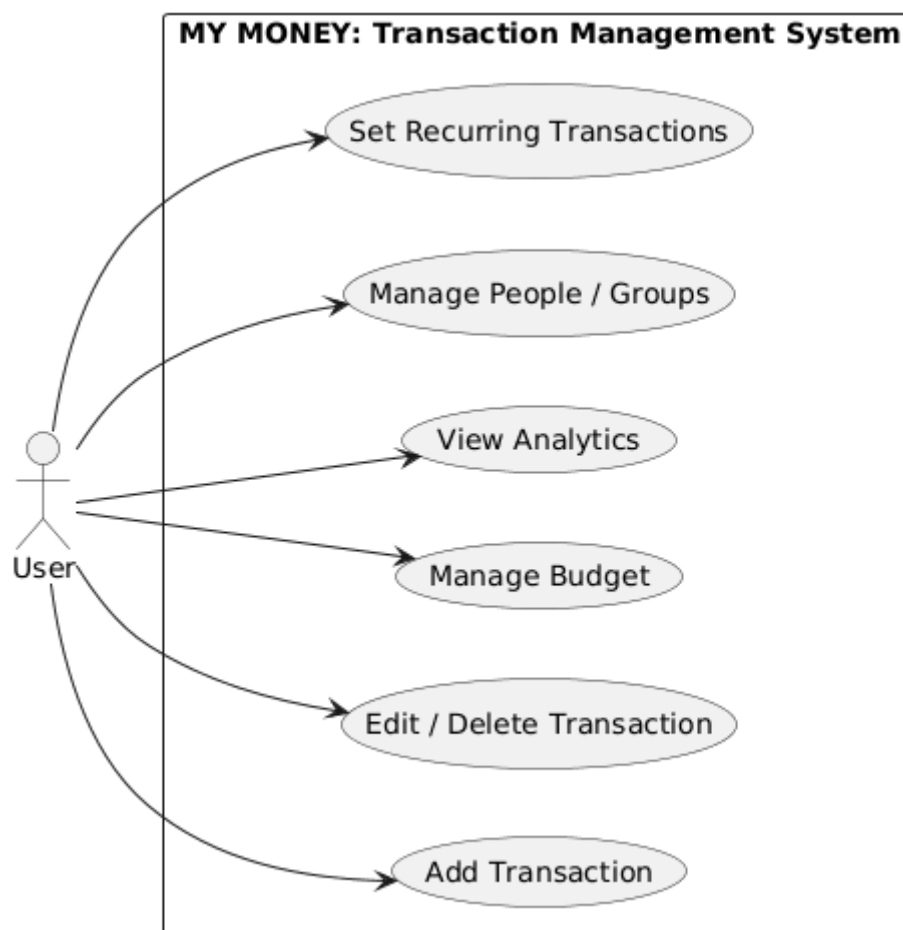


Figure 5.5: Use Case Diagram

5.6 Entity–Relationship Diagram (ERD)

The Entity–Relationship Diagram represents the structure of the database used in the MY MONEY application. It shows the relationship between entities such as transactions, accounts, categories, budgets, people, and groups. This Figure 5.6 helps in understanding how data is organized and how different components are connected within the system.

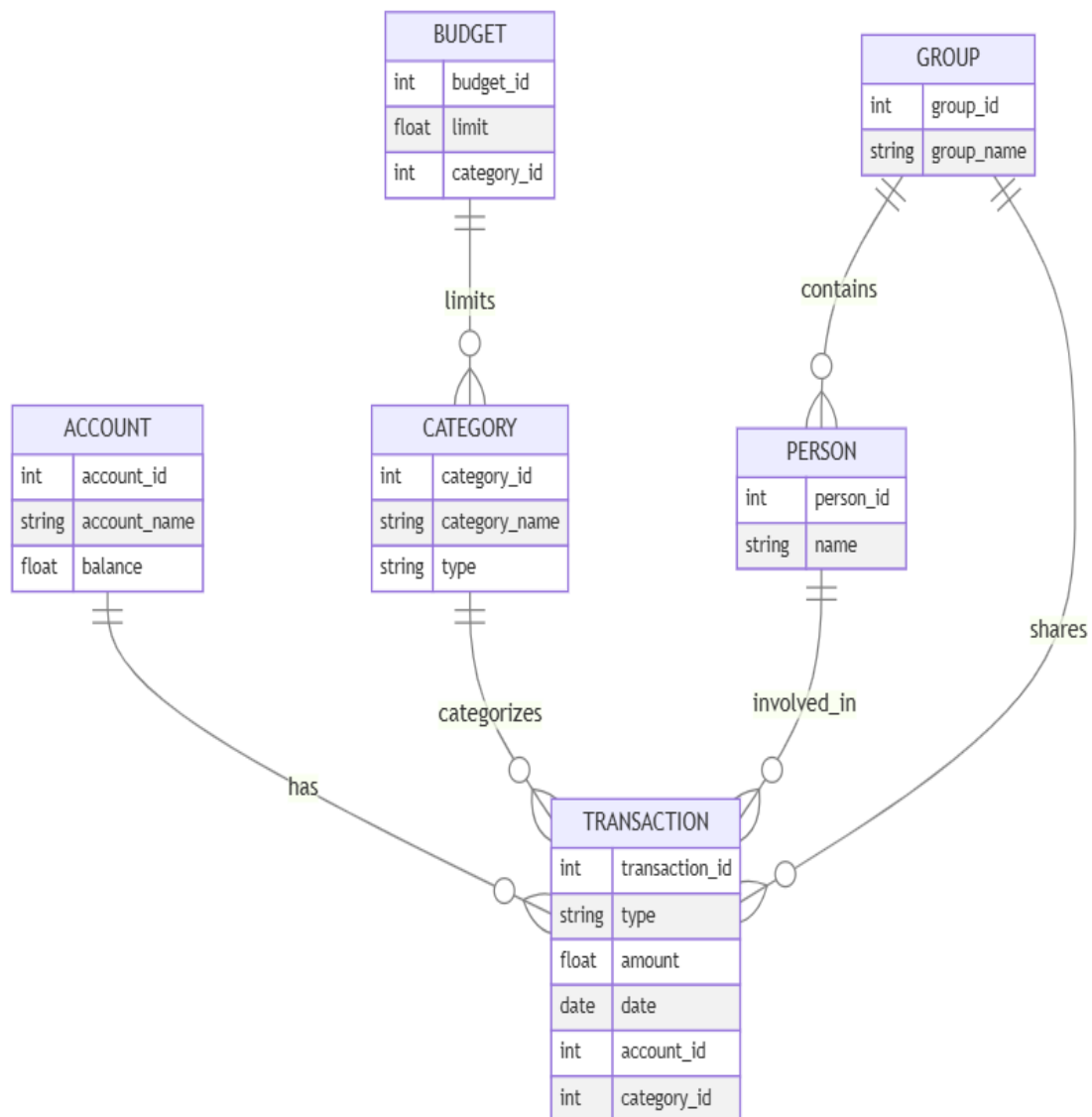


Figure 5.6: Entity–Relationship Diagram

5.7 Activity Diagram

The Activity Diagram represents the sequence of operations performed in the system. It shows the flow of activities such as entering transaction details, validating data, storing information, and updating the system, as illustrated in Figure 5.7.

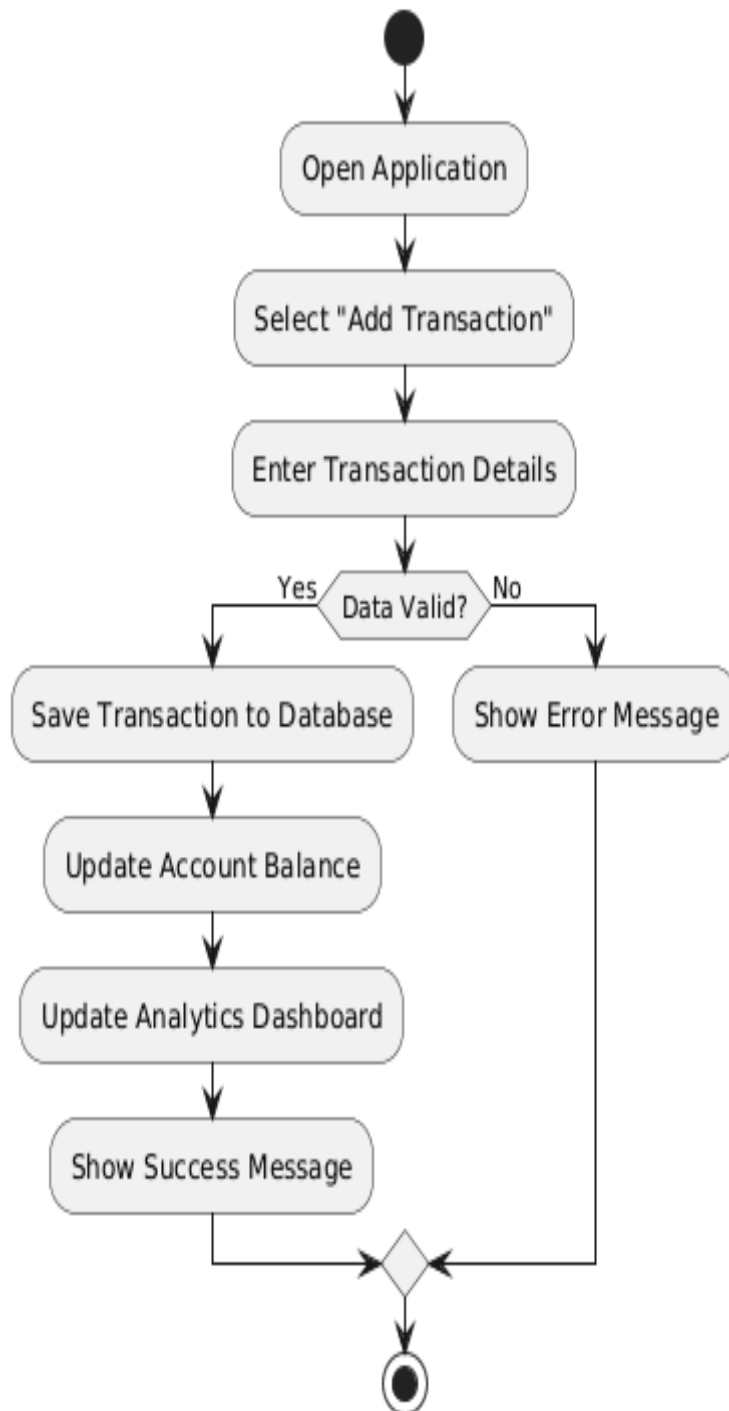


Figure 5.7: Activity Diagram

5.8 Sequence Diagram

The Sequence Diagram shows the interaction between different components of the system over time. It represents how user requests are processed by the application and how responses are generated, as illustrated in Figure 5.8.

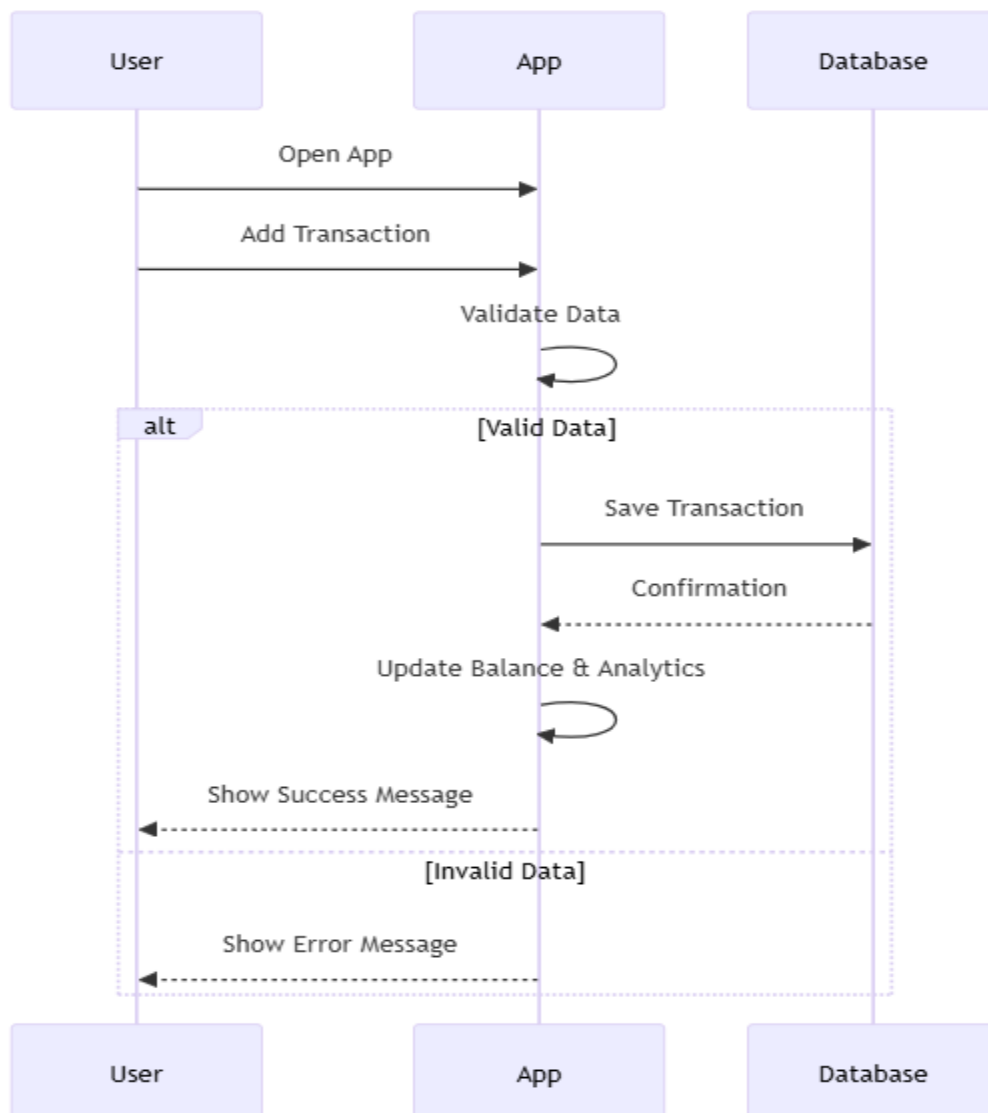


Figure 5.8: Sequence Diagram

5.9 Flowcharts

Flowcharts represent the step-by-step process of different modules in the system. They show how decisions are made and how control flows from one step to another. Important flowcharts include transaction flow, budget management flow, and automation flow, as illustrated in Figure 5.9, Figure 5.10, and Figure 5.11.

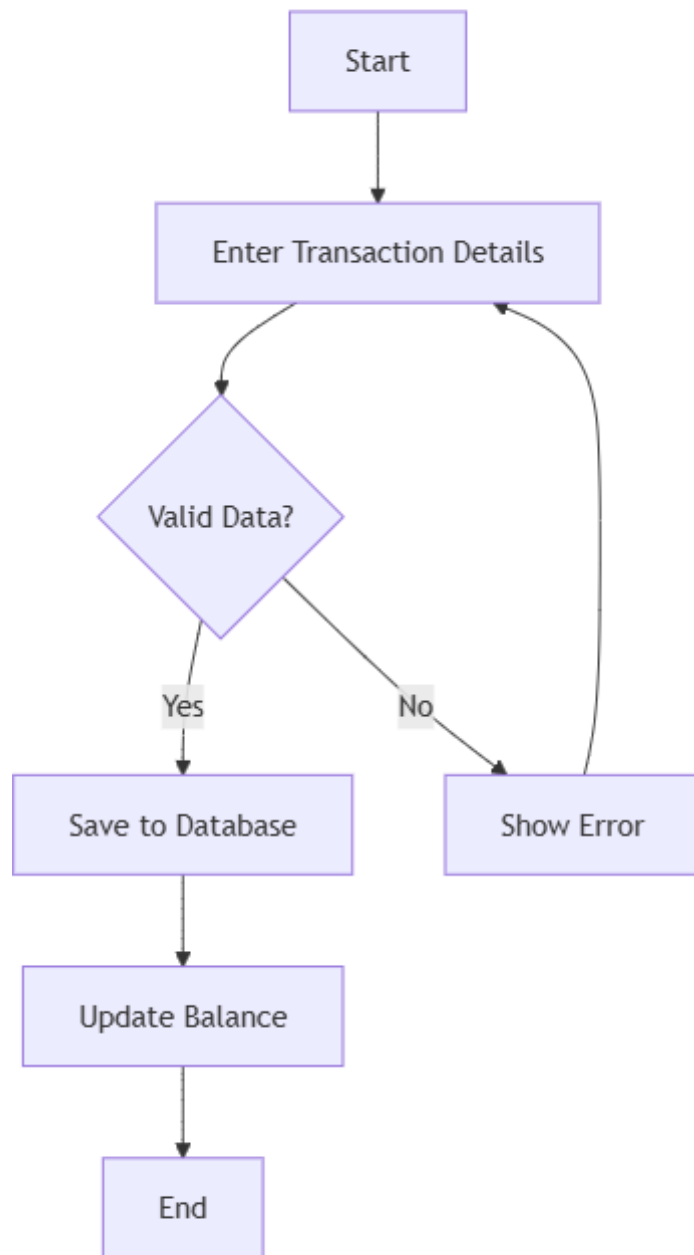


Figure 5.9: Transaction Flowchart

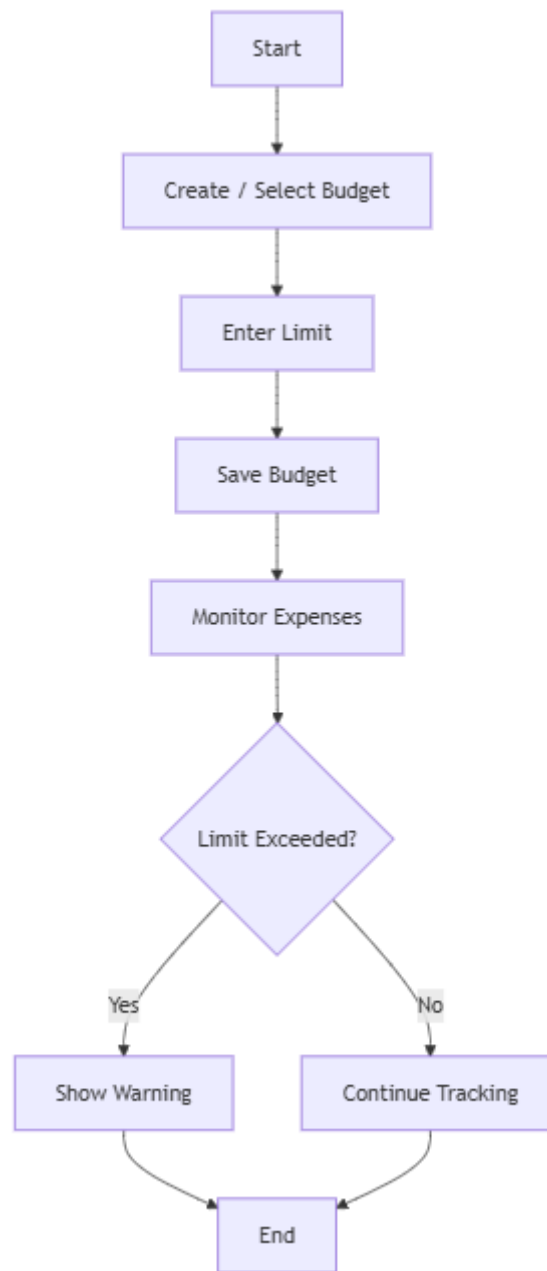


Figure 5.10: Budget Management Flowchart

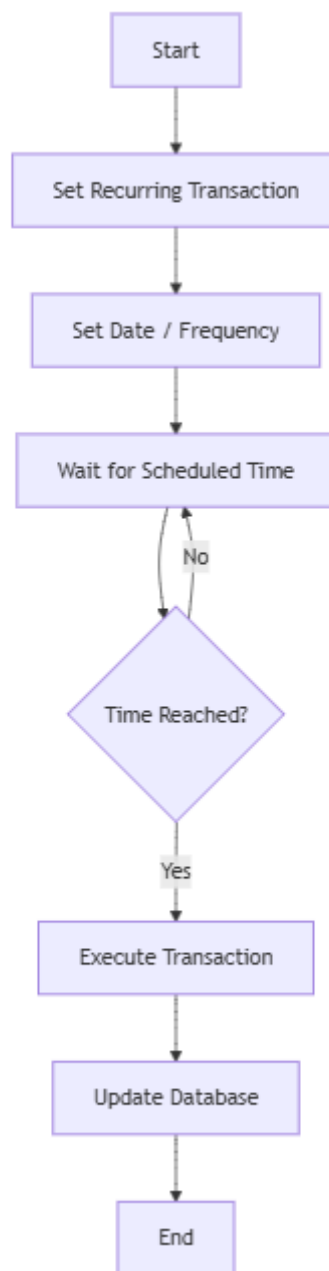


Figure 5.11: Automation Flowchart

This chapter presented the system flow of the MY MONEY: Transaction Management System using various diagrams. These diagrams help in understanding how the system processes data, interacts with users, and manages financial operations. The diagrams provide a clear visualization of system structure, data movement, and interactions between components. The next chapter will focus on the proposed design of the system.

CHAPTER 6

PROPOSED DESIGN

The proposed design of the MY MONEY: Transaction Management System describes how the application is structured and how its components interact to provide a seamless financial management experience. A well-planned design ensures that the system remains efficient, scalable, and easy to maintain. This chapter focuses on the overall design approach, system architecture, user interface planning, navigation structure, API design, and key design considerations. The MY MONEY application is designed as a mobile-based system that operates entirely offline. Therefore, the design emphasizes performance, data security, and smooth user interaction. The system follows a modular approach, where each component is developed independently but works together as part of a unified system.

6.1 Design Approach

The MY MONEY application follows a modular and component-based design approach. Each feature of the system, such as transaction management, budget tracking, relational expense handling, analytics, and automation, is developed as a separate module. This ensures that changes in one module do not affect the overall system. The frontend of the application is built using reusable components such as forms, buttons, lists, and dashboards. This improves consistency and reduces development time. The backend logic is handled locally within the application, where all operations are processed using structured logic and database interactions. The design also ensures that data flows efficiently between different components, allowing real-time updates and smooth user experience.

6.2 System Architecture Design

The system architecture of the MY MONEY application is designed to handle all operations locally within the mobile device. It consists of three main layers that work together to perform all functionalities of the system.

- **Presentation Layer:** This layer includes all user interface components that interact with the user.
- **Application Layer:** This layer handles business logic, data processing, and system operations.
- **Database Layer:** This layer manages data storage using SQLite.

This layered architecture ensures separation of concerns, making the system easier to maintain and extend, as illustrated in Figure 6.1.

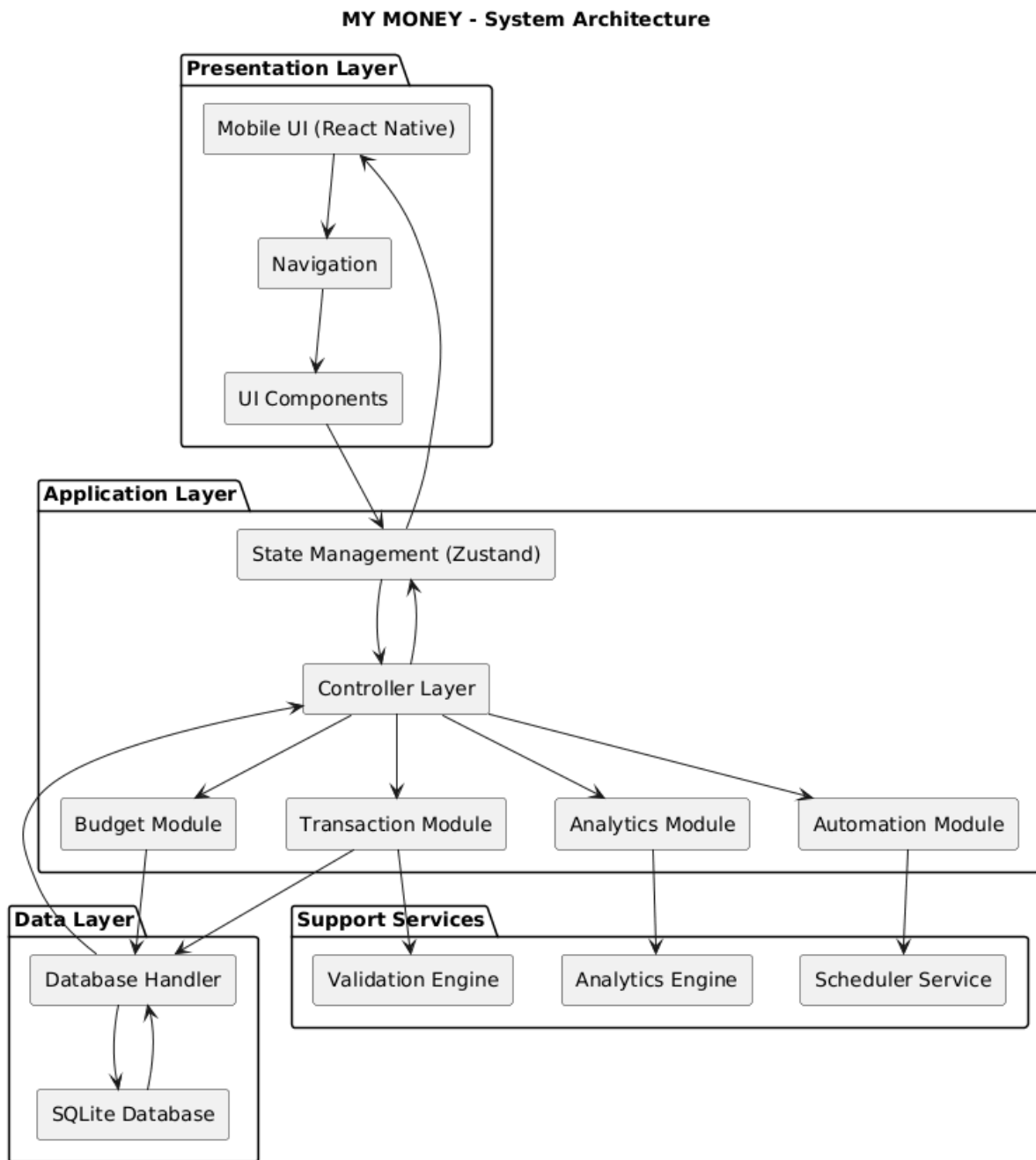


Figure 6.1: System Architecture Diagram

6.3 User Interface (UI) Planning

The user interface of the MY MONEY application is designed to be simple, clean, and intuitive. Since the application is intended for everyday financial management, it is important that users can easily navigate and perform operations without confusion. The UI is designed with consistency in layout, color scheme, and typography. Key screens include transaction entry, dashboard, budget management, and analytics. Each screen is structured to minimize user effort and provide clear information. The dashboard displays a summary of financial data,

while transaction screens allow users to add and manage records easily. The use of visual elements such as charts and graphs enhances user understanding of financial patterns.

6.4 Navigation Design

Navigation design ensures that users can move between different parts of the application easily and efficiently. The MY MONEY application follows a simple and structured navigation flow. The main navigation includes sections such as dashboard, transactions, budgets, analytics, and settings. Users can access these sections with minimal steps, ensuring a smooth experience. Typical navigation flow is as follows:

Home → Dashboard → Add Transaction → View Analytics → Manage Budgets

The navigation is designed to reduce complexity and improve usability, allowing users to perform tasks quickly, as illustrated in Figure 6.2.

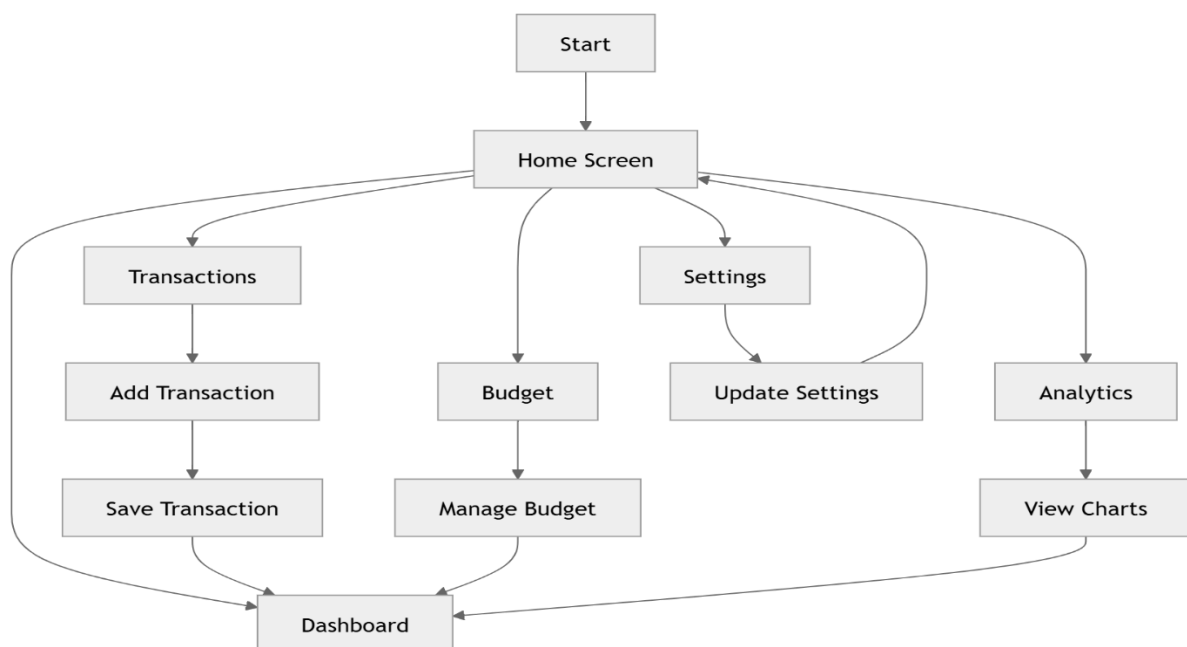


Figure 6.2: Navigation Flow Diagram

6.5 API Design

Since the MY MONEY application operates offline, it does not rely on external APIs. Instead, it uses internal logic and local database operations to handle all functionalities. The system uses structured functions to perform operations such as adding transactions, updating records, retrieving data, and generating analytics. These operations act as internal APIs within the application.

Key functional operations include:

- Transaction operations (add, update, delete)
- Data retrieval for analytics

- Budget and category management
- Automation handling for scheduled transactions

This approach ensures fast performance and eliminates dependency on external services.

6.6 Design Considerations

Several important factors were considered during the design of the MY MONEY application to ensure its effectiveness and reliability.

Performance is a key consideration, as the application must handle large volumes of transaction data efficiently. This is achieved through optimized database queries and efficient data handling.

Security is also an important aspect, as financial data is sensitive. The system ensures that all data is stored locally, reducing the risk of unauthorized access.

Scalability is considered by using a modular architecture, allowing new features to be added easily in the future. Maintainability is ensured through clean code structure and reusable components.

User experience is prioritized by designing an intuitive interface and smooth navigation, ensuring that users can interact with the system easily.

This chapter described the proposed design of the MY MONEY: Transaction Management System. It explained the design approach, system architecture, user interface planning, navigation design, API structure, and key design considerations. The design ensures that the system remains efficient, secure, and user-friendly while supporting future scalability. The next chapter will focus on Experimental Results, which evaluates the performance and functionality of the system.

CHAPTER 7

EXPERIMENTAL RESULT

The experimental results demonstrate the performance, functionality, and reliability of the MY MONEY: Transaction Management System after its implementation. This chapter presents the outcomes observed during testing and validates whether the system meets the requirements defined in earlier chapters. Each module of the application was tested under different scenarios to ensure that it functions correctly and efficiently. The system was tested in a mobile environment using React Native and SQLite as the database. Since the application operates offline, all operations were executed locally on the device. Testing included various user actions such as adding transactions, managing budgets, viewing analytics, and handling relational expense tracking. The results confirm that the system performs effectively under normal usage conditions.

7.1 Overview of Experimental Setup

The MY MONEY application was tested on a standard mobile device with sufficient storage and processing capability. The testing process involved simulating real-world usage scenarios to evaluate system performance and usability.

The testing environment included:

- Mobile device (Android/iOS)
- Local SQLite database
- Application built using React Native and Expo

Different test cases were executed to check the behaviour of each module, including transaction handling, data storage, analytics generation, and automation features.

7.2 Transaction Management Results

The transaction module was tested to verify whether users can add, edit, and delete financial records correctly. The system successfully recorded all types of transactions, including income, expenses, and transfers. The results of this testing are presented in Table 7.1. The screen of the transaction Creation is illustrated at Figure 7.1.

Table 7.1 Transaction Module Results

Test Scenario	Expected Output	Result
Add transaction	Transaction stored successfully	Successful
Edit transaction	Updated data reflected	Successful
Delete transaction	Record removed from database	Successful
Multiple entries	All entries stored correctly	Successful

The results confirm that the transaction module functions accurately and maintains data consistency.

Figure 7.1: Add Record Screen

7.3 Budget Management Results

The budget module was tested to ensure that users can create and manage categories effectively. The system successfully grouped transactions under different categories and maintained proper budget tracking. Users were able to assign categories, update them, and view categorized financial data without any issues. The module performed efficiently even with multiple categories. The Budget Creation is shown at Figure 7.2.

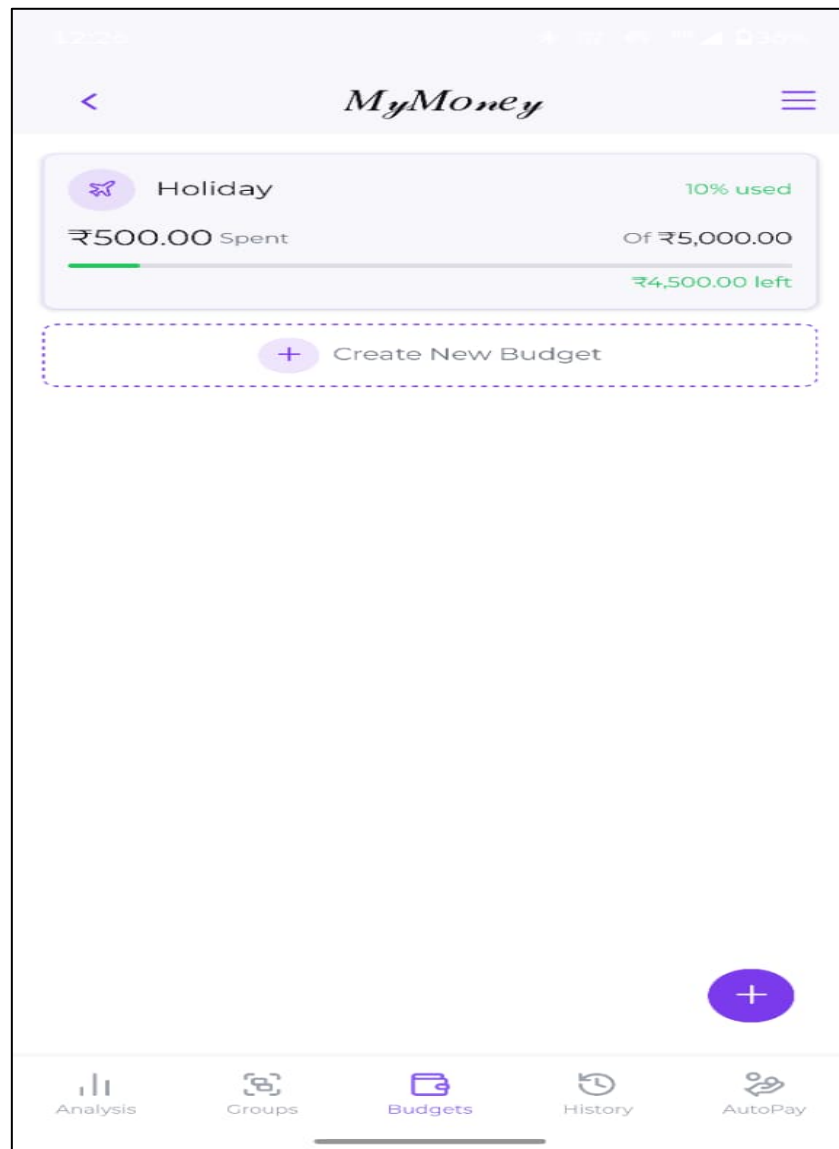


Figure 7.2: Budget Screen

7.4 Relational Tracking Results

The relational tracking module was tested for handling shared expenses among individuals and groups. The system successfully calculated and distributed expenses based on user input. The results showed that shared transactions were recorded correctly and relationships between users and expenses were maintained properly. This feature improves the overall functionality of the application by providing detailed financial tracking.

7.5 Analytics and Dashboard Results

The analytics module was tested to verify whether financial data is displayed correctly through charts and summaries. The system successfully generated graphical representations based on stored data. The results of this testing are presented in Table 7.2. The Analysis screen is shown at Figure 7.3.

Table 7.2 Analytics Results

Feature	Observation
Data accuracy	Correct financial calculations
Chart display	Proper visualization
Real-time updates	Instant updates after transactions
Performance	Smooth rendering

The dashboard provided clear insights into spending patterns, allowing users to understand their financial status easily.

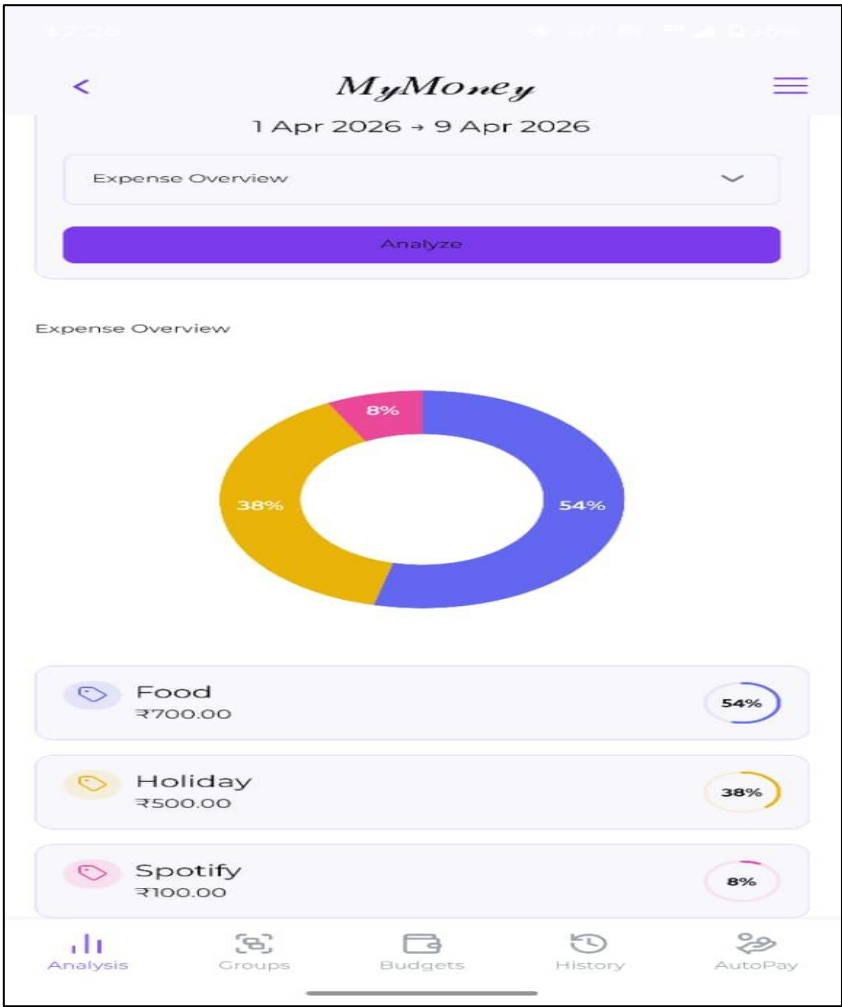


Figure 7.3: Expense Analysis

7.6 Automation Module Results

The automation module was tested to evaluate scheduled and recurring transactions. The system successfully executed predefined transactions without requiring manual input. The results confirm that the automation feature works efficiently and reduces user effort in managing repeated financial activities. The autopay screen is illustrated at Figure 7.4.

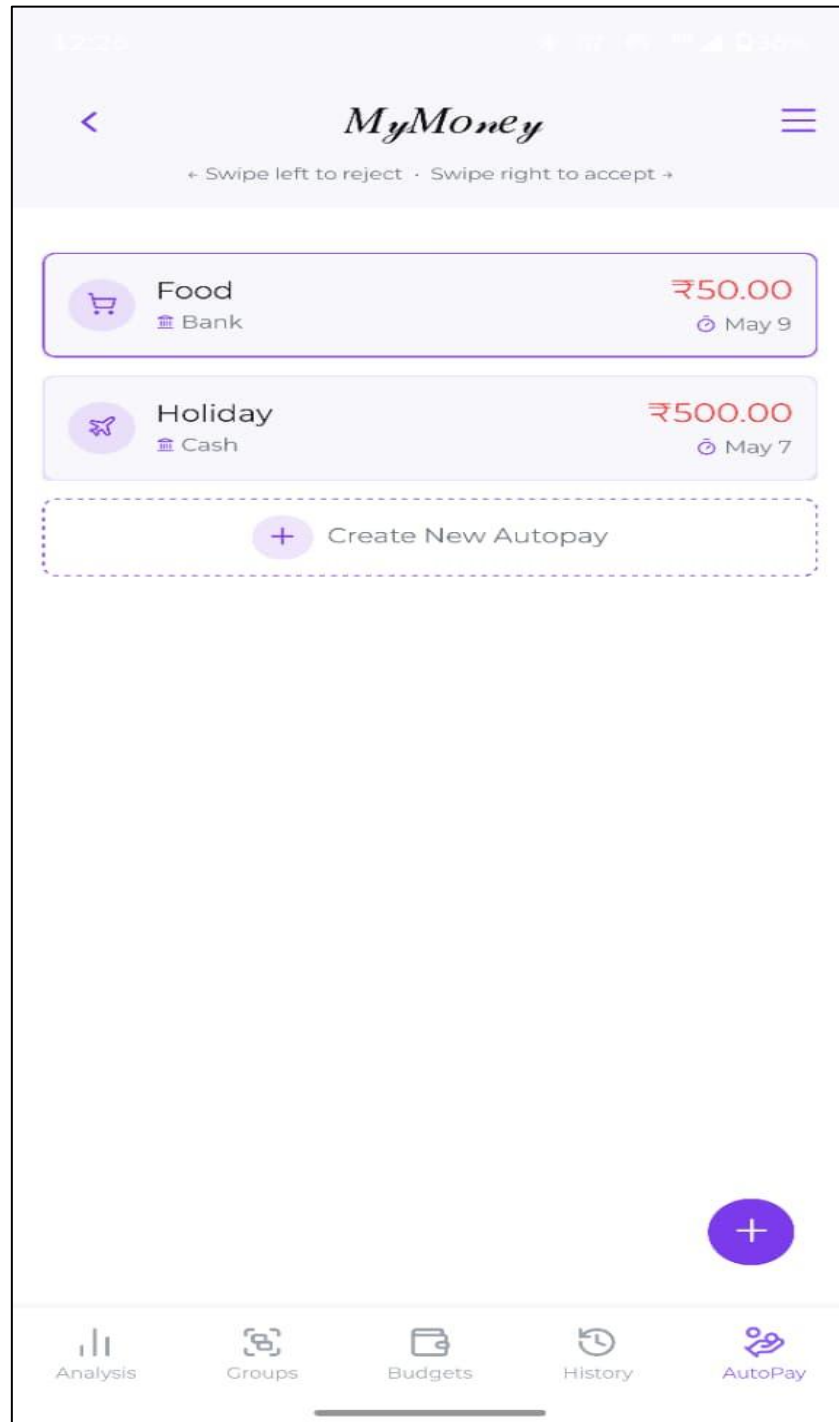


Figure 7.4: AutoPay Screen

7.7 Performance Observations

The overall performance of the MY MONEY application was observed to be stable and efficient. Since all operations are performed locally, the system provides fast response times and smooth user interaction.

Key observations include:

- Quick data processing and retrieval
- No dependency on internet connectivity
- Smooth navigation across all modules
- Efficient handling of large data sets

The use of SQLite ensures that the system remains responsive even with a large number of transactions.

7.8 Testimonials

The MY MONEY: Transaction Management System was evaluated by a group of users to assess its usability, performance, and overall effectiveness. The feedback collected highlights the practical benefits of the application and its ability to simplify financial management tasks.

“The application is very easy to use, and the interface is simple and intuitive. Managing transactions is smooth and fast.” – Sourindro Biswas (Date: 12 April 2026)

“I really appreciate the offline functionality. It allows me to manage my financial data without depending on internet connectivity.” – Sumit Das (Date: 14 April 2026)

“The analytics feature provides clear insights into my spending habits through well-presented graphs.” – Pritam Ghosh (Date: 16 April 2026)

“The relational tracking feature is very useful for managing shared expenses among friends and groups.” – Sanjib Barik (Date: 18 April 2026)

“The automation feature helps in handling recurring transactions efficiently and reduces manual effort.” – Arjun Ghosh (Date: 20 April 2026)

Some users also suggested improvements such as cloud synchronization, advanced reporting features, and additional customization options. These enhancements can be considered for future development of the system. Overall, the feedback indicates that the MY MONEY application is user-friendly, efficient, and reliable, successfully meeting its intended objectives.

CHAPTER 8

FUTURE SCOPE

The MY MONEY: Transaction Management System has been designed as a comprehensive and efficient financial management solution. However, there are several opportunities for further enhancement and expansion to improve its functionality and user experience.

One of the major future enhancements of the system is the integration of UPI-based payment functionality. Currently, users need to manually record transactions after making payments through external applications. The proposed integration will allow users to initiate and complete payments directly within the application, with automatic transaction recording. This will eliminate manual effort and ensure real-time synchronization between financial activities and recorded data.

Another important area of improvement is cloud synchronization. At present, the system follows an offline-first approach, where all data is stored locally on the device. In the future, cloud integration can be introduced to enable multi-device access and secure data backup. This will allow users to access their financial data across different devices while maintaining data consistency.

The application can also be enhanced by introducing advanced expense management features such as percentage-based and fractional expense splitting. This will improve the flexibility of group expense management and reduce manual calculations.

Integration with banking systems through secure APIs is another potential improvement. This will allow automatic fetching of transaction data from bank accounts, reducing the need for manual entry and improving accuracy.

Furthermore, the system can incorporate intelligent features such as an AI-based financial assistant. This assistant can analyze user data and provide insights, spending trends, and personalized financial suggestions.

Additional features such as receipt scanning using OCR (Optical Character Recognition) can also be implemented. This will allow users to capture expense details directly from bills and receipts, making data entry faster and more convenient.

Finally, the system can be expanded to support multi-currency transactions, enabling its use in global financial environments. This will make the application more scalable and suitable for a wider range of users.

Overall, these enhancements will improve the system's usability, scalability, and intelligence, making MY MONEY a more powerful and complete financial management solution.

CHAPTER 9

CONCLUSION

The MY MONEY: Transaction Management System is not merely an incremental improvement over existing financial applications, but a comprehensive rethinking of personal finance management. It is based on the idea that expense tracking, budget management, social debt handling, and recurring payment management are interconnected aspects of a single problem, which should be addressed through an integrated solution rather than separate applications.

The system has been designed using a robust and reliable technical architecture. The use of SQLite ensures data consistency and integrity, while efficient state management enables smooth performance. The application maintains accurate transaction records and supports structured financial tracking, making it dependable for real-world usage.

In terms of functionality, the system provides a complete solution for managing personal finances. It includes features such as a unified transaction ledger, peer-to-peer debt tracking, group expense management, budget monitoring, and automation of recurring transactions. These features address practical financial needs that are often only partially supported by existing applications.

The MY MONEY system also demonstrates a strong competitive advantage. Compared to existing applications, it offers better data accuracy, enhanced privacy, and a more user-friendly approach. Its offline-first design ensures that users can access and manage their data without depending on internet connectivity, while local data storage enhances security and user trust.

The future scope of the system further strengthens its potential. Planned features such as UPI payment integration, cloud synchronization, AI-based financial insights, and receipt scanning will enhance both usability and intelligence. These improvements will transform the system from a financial tracking tool into a complete financial management platform.

Finally, the most significant aspect of the MY MONEY system is its strong emphasis on privacy. All financial data is stored locally on the user's device, ensuring complete control and security. This privacy-first approach forms the foundation of the system and builds trust among users.

In conclusion, the MY MONEY: Transaction Management System is a reliable, efficient, and innovative solution that successfully addresses the complexities of personal finance management. With its strong architecture, comprehensive features, and clear future roadmap, it has the potential to become a highly effective tool for users seeking better financial control and transparency.

APPENDICES

Appendix A: User Interface Screens

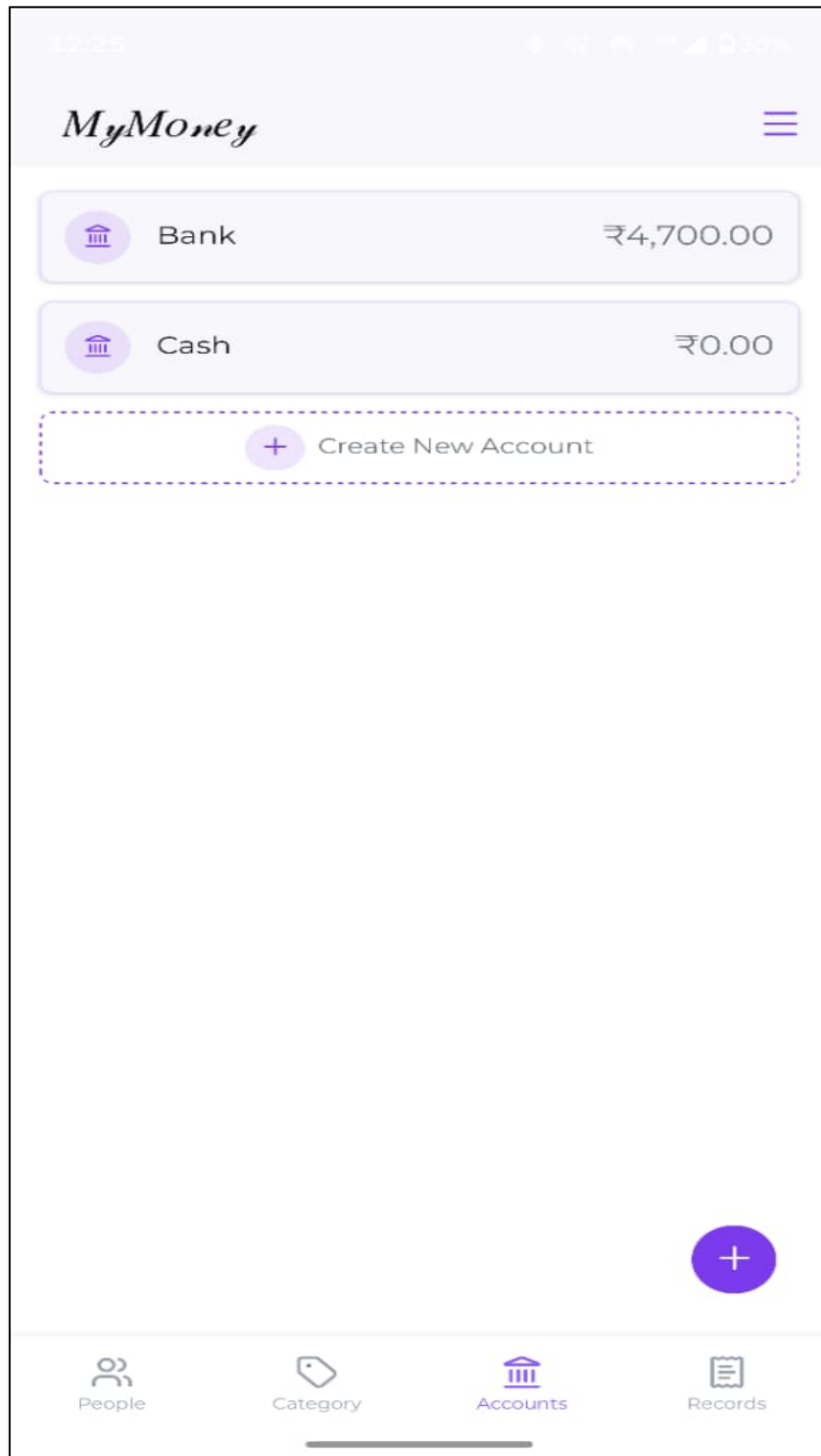


Figure A.1: Accounts Screen

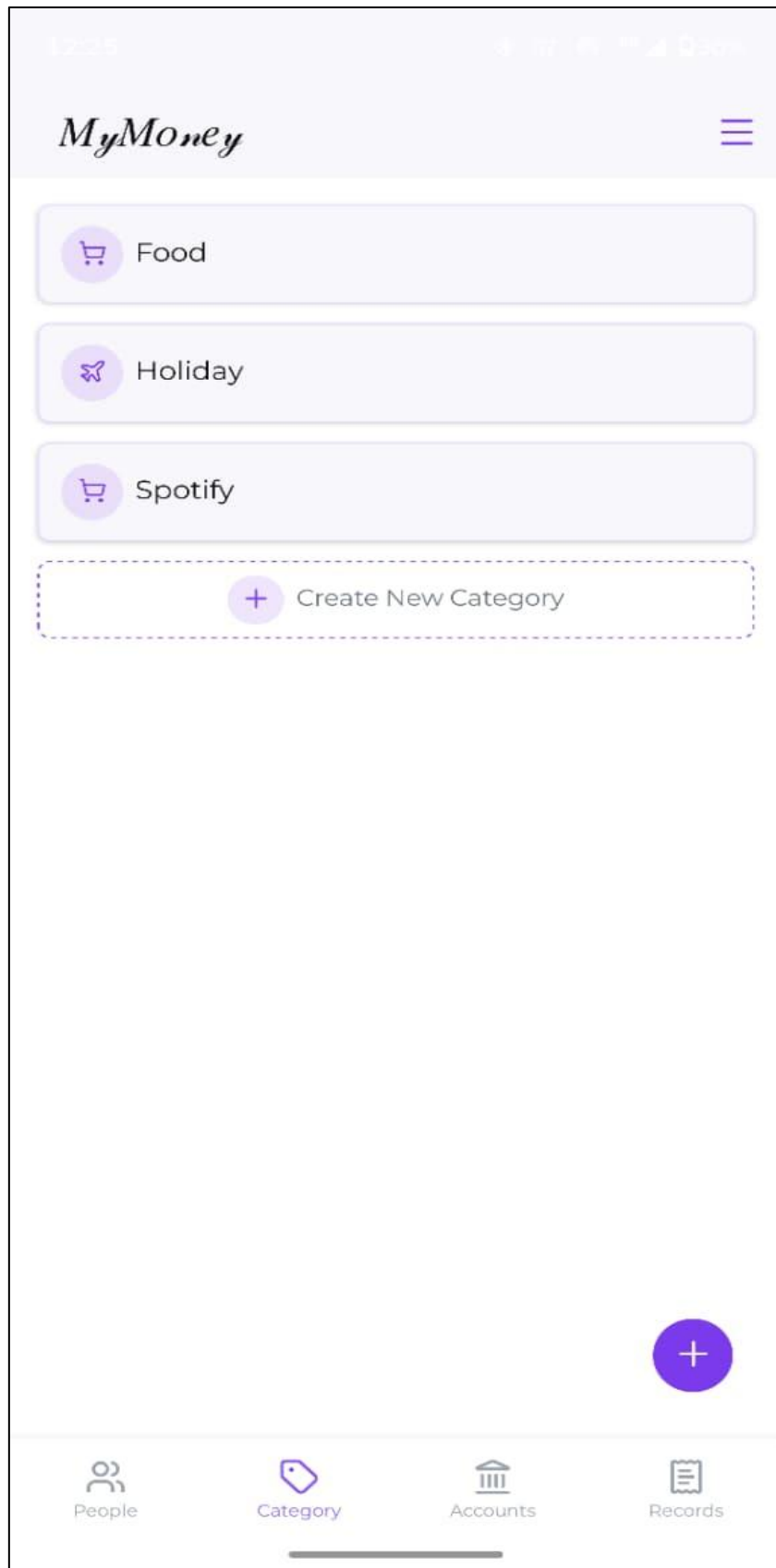


Figure A.2: Categories Screen

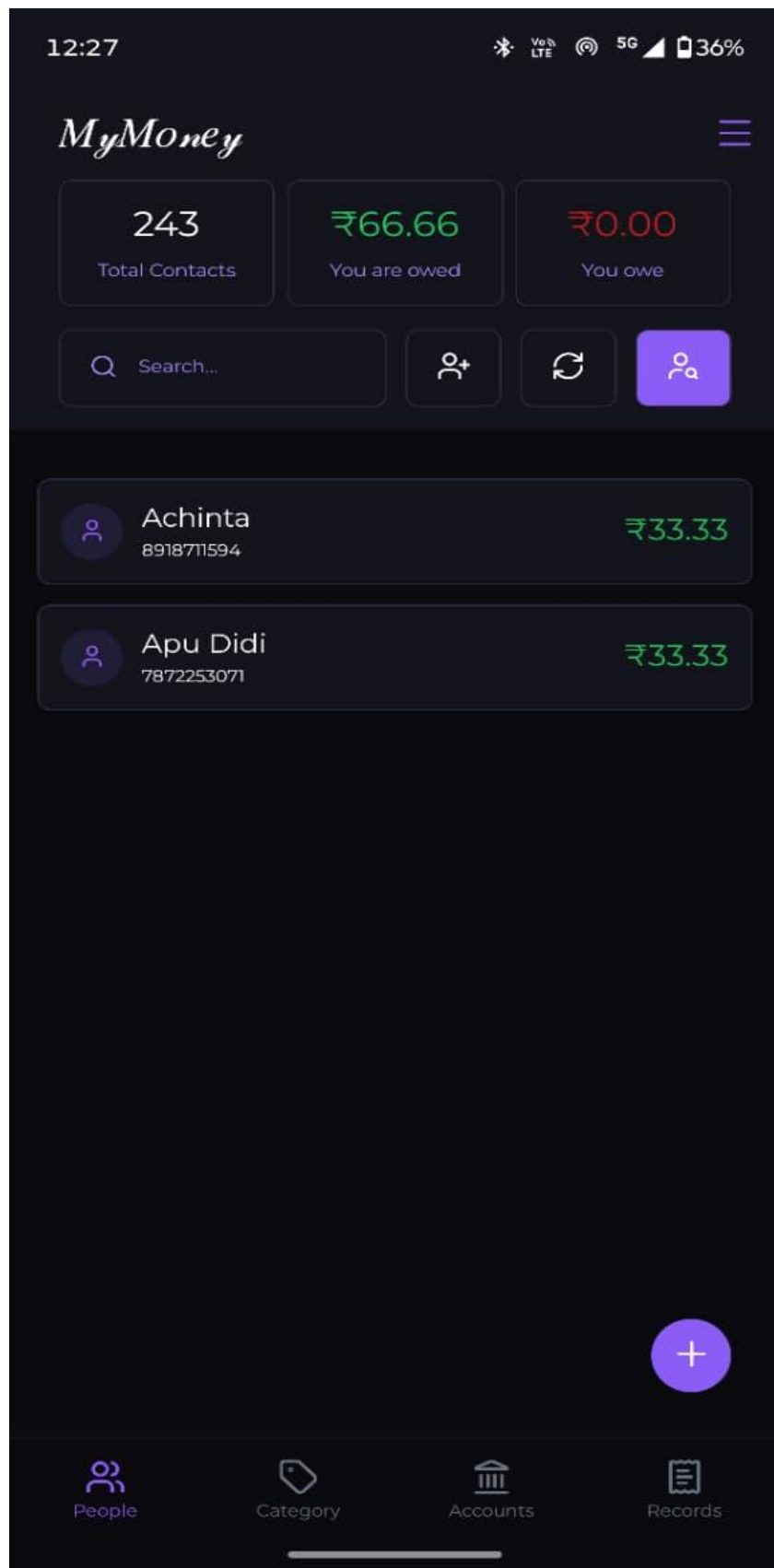


Figure A.3: Dark Mode UI

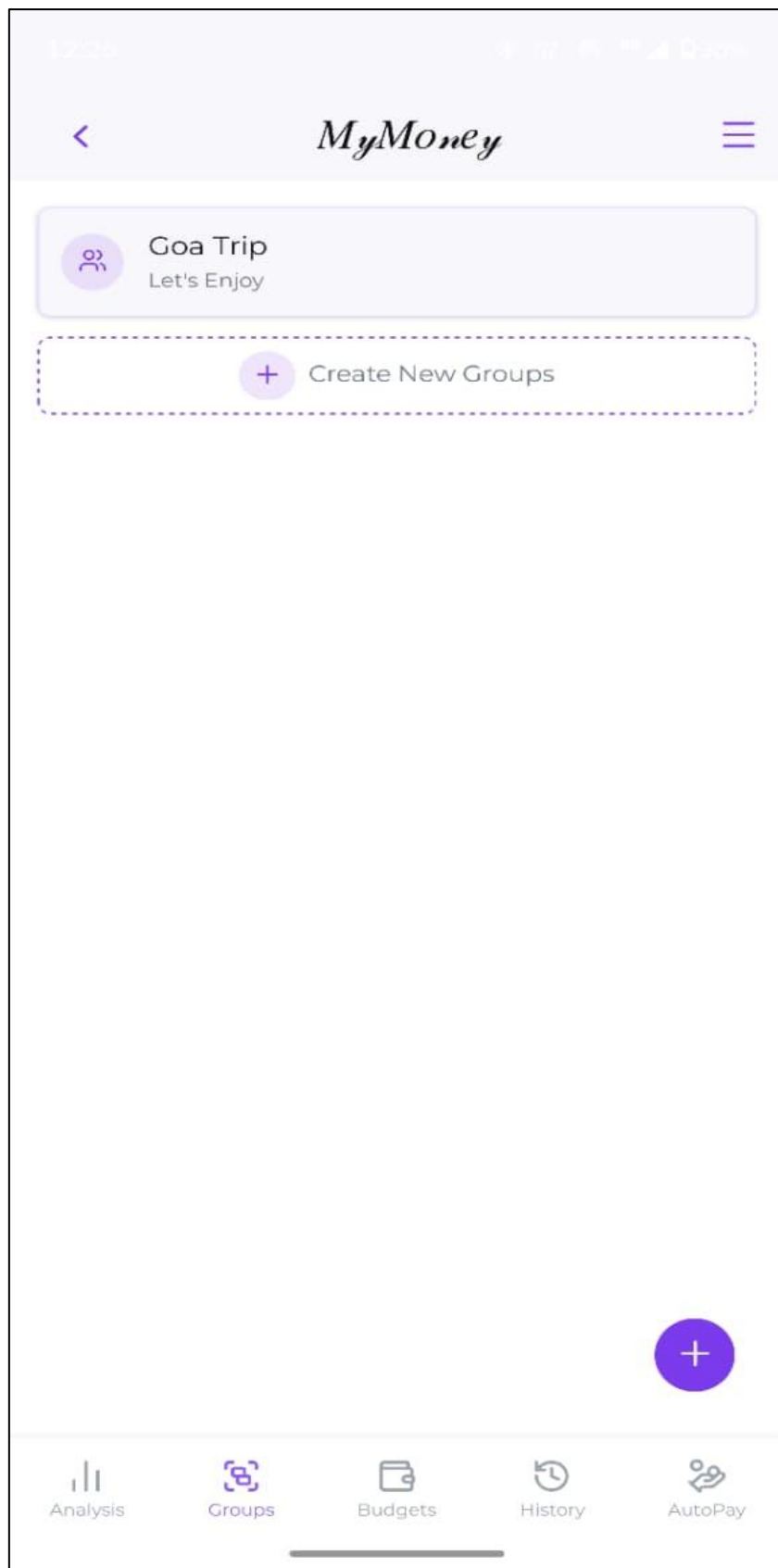


Figure A.4: Groups Screen

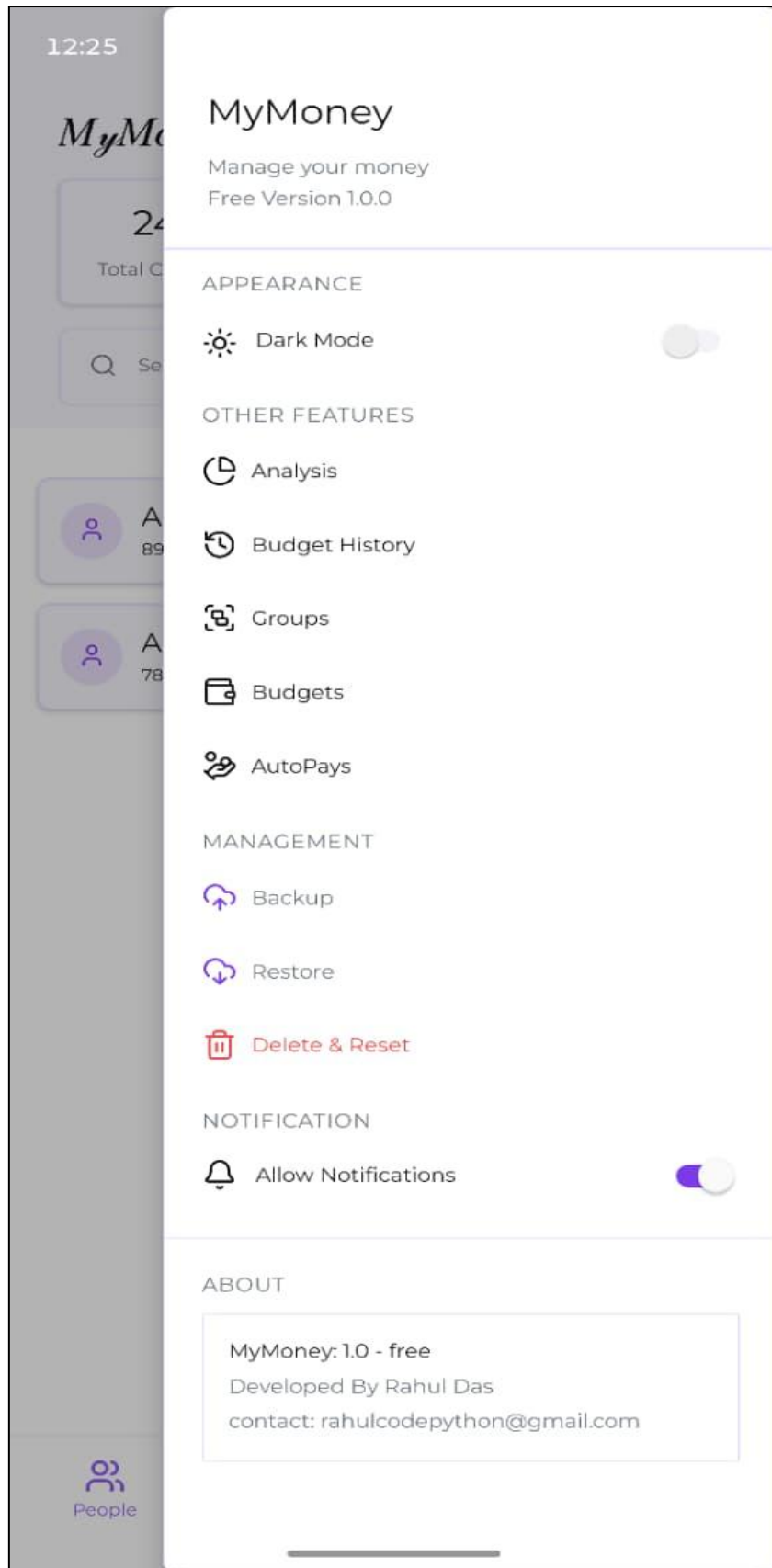


Figure A.5: Navigation Menu

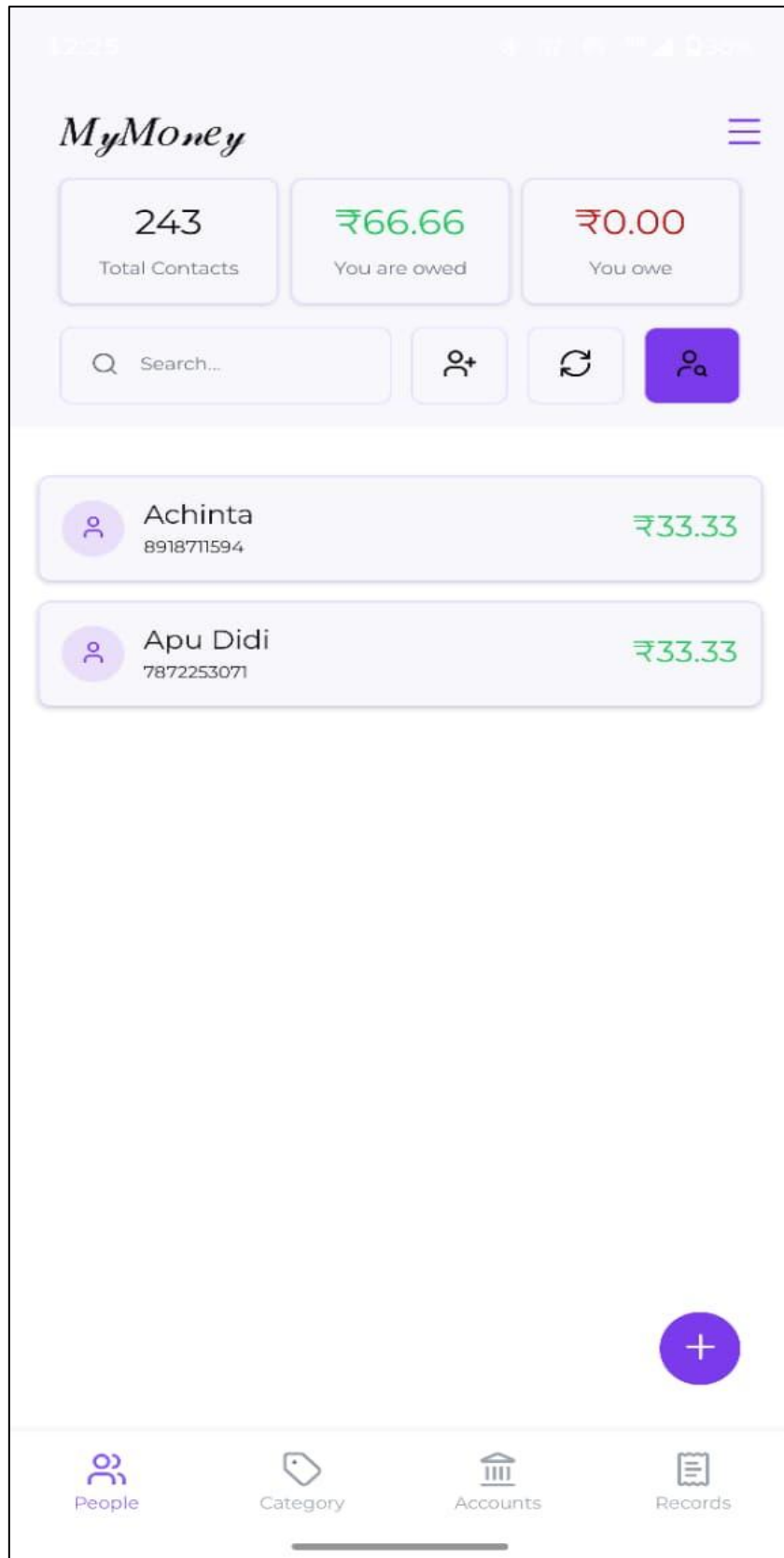


Figure A.6: People & Debts Screen

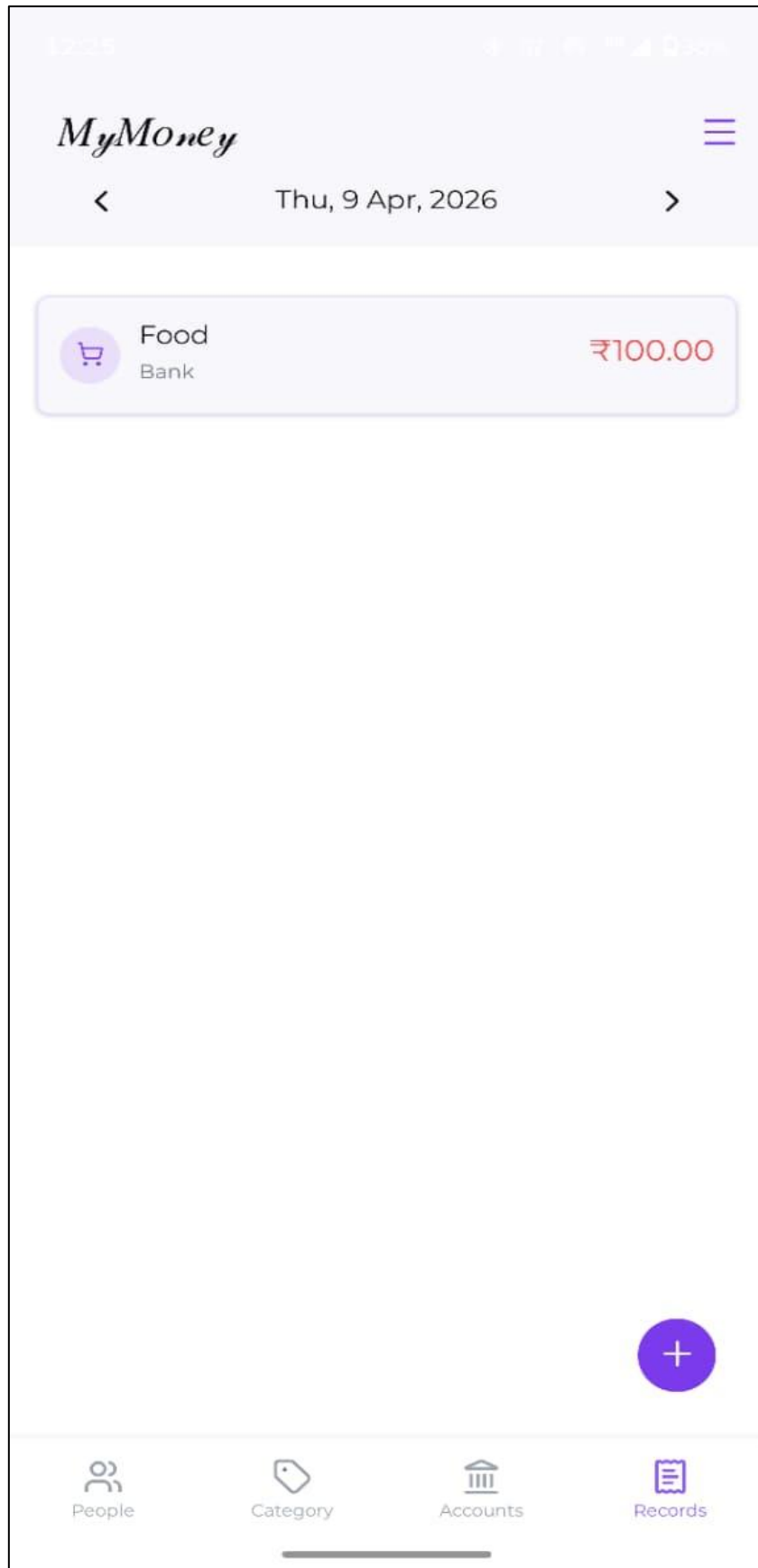


Figure A.7: Daily Records Screen

Appendix B: Sample Code

Code B.1 SQLite Table Creation (Transactions Table)

```
CREATE TABLE transactions (  
  id TEXT PRIMARY KEY,  
  account_id TEXT,  
  category_id TEXT,  
  amount REAL,  
  type TEXT,  
  date TEXT,  
  start_balance REAL,  
  end_balance REAL  
);
```

Code B.2 Insert Transaction Query

```
INSERT INTO transactions  
(id, account_id, category_id, amount, type, date, start_balance, end_balance)  
VALUES  
('uuid-123', 'acc-001', 'cat-001', 500, 'expense', '2026-03-20', 10000, 9500);
```

Code B.3 Zustand State Management (Basic Store)

```
function processAutopay(autopayList) {  
  const today = new Date();  
  autopayList.forEach((item) => {  
    if (new Date(item.next_due_date) <= today) {  
      addTransaction({  
        amount: item.amount,  
        category_id: item.category_id,  
        account_id: item.account_id,  
        type: "expense",  

```

```

        date: today.toISOString()
    });
}
});
}

```

Code B.4 Recurring Transaction Logic (Autopay Example)

```

function processAutopay(autopayList) {
    const today = new Date();
    autopayList.forEach((item) => {
        if (new Date(item.next_due_date) <= today) {
            addTransaction({
                amount: item.amount,
                category_id: item.category_id,
                account_id: item.account_id,
                type: "expense",
                date: today.toISOString()
            });
        }
    });
}

```

Appendix C: Database Structure

Table C.1 Database Tables Overview

Table Name	Purpose	Key Fields	Relationships
accounts	Stores all financial containers (bank, wallet, cash, credit)	id, name, initial_balance, icon	Referenced by transactions, transfers
categories	Defines income/expense classification	id, name, type, color, icon	Referenced by transactions, budgets

transactions	Core financial ledger entries	id, account_id, category_id, amount, type, date, start_balance, end_balance	Links to accounts, categories, transactions_people, transaction_group
transfers	Internal fund movement between accounts	id, from_account_id, to_account_id, amount, date	Links to accounts
people	Stores contacts for debt tracking	id, name	Referenced by transactions_people
transactions_people	Junction table for debt engine	id, transaction_id, person_id, balance_owed	Links transactions ↔ people
groups	Logical clustering of transactions	id, name, created_at	Referenced by transaction_group
transaction_group	Junction table for grouping	id, transaction_id, group_id	Links transactions ↔ groups
budgets	Monthly category budgets	id, category_id, limit_amount	Links to categories
budget_history	Archived monthly budget performance	id, category_id, month, spent, limit	Historical snapshot
autopay	Recurring payment templates	id, account_id, category_id, amount, frequency, next_due_date	Links to accounts, categories

Appendix D: Test Cases

Table D.1 Test Cases for MY MONEY System

Test Case ID	Module	Test Description	Input	Expected Output	Actual Output	Result
TC-01	Transaction	Add new transaction	₹500 expense	Transaction saved successfully	Transaction saved	Pass
TC-02	Transaction	Edit transaction	Change ₹500 → ₹600	Updated transaction	Updated successfully	Pass
TC-03	Transaction	Delete transaction	Delete entry	Transaction removed	Removed successfully	Pass
TC-04	Budget	Create budget	Limit ₹2000	Budget created	Created successfully	Pass

TC-05	Budget	Exceed budget	Expense > limit	Warning message	Warning shown	Pass
TC-06	Analytics	View analytics	Open dashboard	Charts displayed	Charts shown	Pass
TC-07	Relational	Add group expense	₹1000 split	Expense shared	Shared correctly	Pass
TC-08	Automation	Set recurring transaction	Monthly ₹500	Auto transaction added	Executed correctly	Pass
TC-09	Validation	Invalid input	Empty amount	Error message	Error shown	Pass
TC-10	Database	Store data	Add entry	Data stored	Stored correctly	Pass

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